Matching a Physical Drum Model to a Recording

A perceptual distance in readable units, and what it establishes about a modal tom

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Abstract. Fitting a physical instrument model to a recording needs an objective that a person can read. This paper describes one: a recording and a render are reduced to the same set of perceptually named quantities — partial frequencies and per-partial ring times, a time-varying fractional-octave envelope, an amplitude envelope, pitch glide, attack balance, and coverage in both directions — and each term of the distance is reported in its own unit, so that a result can be read against a tolerance rather than only against another run. Adoption is decided per term. The reference reduction measures inter-channel delay and aligns before averaging, which on this recording decides which partials survive detection at all. The search is deterministic given its seed, checkpointed, and refuses to resume across a change in the model or the measurement.

The instrument the method is applied to is described here in full, because every conclusion the fit reaches is a claim about it: two membranes coupled through a short modal expansion of the enclosed air, a Berger tension nonlinearity closed by a discrete gradient and extended by the mode-to-mode channels Berger projects away, a three-term loss law shaped to hold constant Q, a microphone model that is a Lommel integral plus an evanescent term, and a stochastic layer whose releases are read off that same loss law past the modes the budget can afford. Which of its numbers are derived, which are measured and which are fitted is reported alongside them.

Applied to that tom, the apparatus takes the distance from 33.1 at the shipped default to 11.3, bringing five of the nine terms below their thresholds — amplitude envelope, glide, attack balance and both coverage shares — and it localises what remains. What remains is a **shape** the loss law cannot make: the model delivers a nearly flat 0.6–1.0 s ring across 370–700 Hz where the reference scatters over a factor of four, and rings about three-quarters as long as the reference at its fundamental. The head-damping parameter, which scales all loss rates together, was fitted against the lower bound of its range in an earlier run. That is a specific, testable statement, and it is put to the test here: given four times the head-damping headroom through a build-time multiplier that leaves the shipped parameter range untouched, the search chooses the same physical damping to within 2%, so the residual is a question about the shape of the loss law rather than about its range. The same per-term reading separates two contact models fitted at identical budget — 11.252 against 11.535, winning on different terms, by a margin Section 12 then puts back in question — and a second experiment exploits the fact that mode frequencies are analytic: a pre-solve seeds restarts with banks whose modes already lie on the reference's partials, worth 12% at identical cost against a target it reaches to a cent, and worth nothing against one it reaches only to thirty-six — a distinction the pre-solve itself reports before the search begins. A final chapter refits the prescribed model under the corrected glide estimator and reaches 10.382 from a 32.442 baseline; a second, independent search agrees with it on every term the objective can read, which places the residual at the model's ceiling on this recording rather than at the search's.

Working-paper status. The fitted numbers in Section 8 come from two complete reports, one per contact model, each a full-budget run of eight restarts under the corrected reduction of Section 11; the figures and every number in that section are read from the better of the two. Section 9 and Section 10 report earlier runs at their own stated budgets, each paired against a control measured at the same budget.

Those earlier runs predate the correction to the partial-level estimator reported in Section 11 (2026-07-31), which changed the reference's partial list and every candidate's. Their totals and term values are therefore **not comparable to anything measured after that date**, and are reported here for the method they establish rather than as current results; Section 11 reports the correction and what it re-scopes. Under the corrected reduction the shipped bank measures 33.094 with prescribed contact and 33.544 with Hertzian, at the Standard tier against the right channel.

Every fit up to Section 12, including those two, predates a second correction: the glide estimator was faulty in both of its probes (Section 12, 2026-07-31). That term sat inside the objective the search minimised, so the fitted banks are conditioned on it and not merely mis-scored by it. Section 8 and Table 9 are reported as obtained rather than re-run, and Section 12 gives the fault, its size, the corrected reference figure and what specifically it puts back in question.

Section 13 is the one run made **after** that correction (2026-08-01) and the only section whose total is current: the prescribed contact refitted from scratch, 32.442 → 10.382. It also re-measures four archived banks under the corrected objective so that they can be read on one axis. The Hertzian contact has not been refitted, so nothing in this paper establishes the contact-model ordering.

1. THE PROBLEM

Fitting a physically parameterised synthesiser to a recording is an optimisation whose objective is the whole difficulty. A waveform difference is unusable against a recording of a different physical instrument. A spectral distance is usable but opaque: it produces one number, and when that number stops improving there is nothing to say about what is still wrong.

The apparatus described here takes the opposite approach. Every term is a quantity a listener would name, in the unit that quantity is perceived in, and the sum exists only to order candidates — decisions are made term by term. The result is that a fit does not merely converge; it says what it has achieved and what it has not, and the latter turns into statements about the model.

2. THE INSTRUMENT

The instrument is a double-headed tom rendered by modal synthesis: two membranes coupled through a modal expansion of the enclosed air, with a Berger tension nonlinearity and its mode-coupling extension, a microphone model, and a three-band stochastic attack layer covering the transient region modal synthesis reaches poorly. Its configuration is SI-valued and versioned; the product exposes eighteen normalised parameters over it.

This chapter describes it in enough detail to argue with. That is not scene-setting. Every conclusion the rest of the paper reaches is a statement **about this model** — that its head-damping bound is not binding, that the residual lives in the shape of its loss law rather than in the range of any parameter, that its mode frequencies are analytic and therefore cheap enough to seed a search from — and none of those can be checked, or disputed, from a description that stops at the word "modal".

2.1. Two membranes, an air cavity, and a microphone

The signal path is short and its asymmetries are deliberate. A strike deposits force at one point on the **batter** head, spread over a finite contact footprint. The batter head's modes are the only ones the stick reaches. The **resonant** head is driven solely by the enclosed air, which is expanded in the rigid-walled cylinder's own modes — six pressure states at the shipped default, of which the uniform one is the lumped spring compressed by the net volume the two heads sweep. Each head's total strain raises its own tension, which detunes every one of its modes together, and the part of that quartic potential the tension law projects away is restored as a small set of mode-to-mode channels, so a loud hit also deposits energy at frequencies nothing struck. A stochastic layer, driven by the same contact force, supplies the band above the highest mode the budget can afford. What the listener hears is a weighted sum of the batter head's modal accelerations and that layer, band-limited and scaled.

Two properties of that path are worth naming immediately, because they are easy to assume otherwise. Only the batter head radiates into the output: the resonant head is fully coupled into the dynamics but its own radiation leaves the far side of the shell, and adding it at the same point, phase and distance would be a fiction. And the cavity reaches only the azimuthal orders its own basis carries — $m = 0$ alone when the air is one lumped state, because the uniform pressure responds to swept volume and every other mode sweeps exactly none, and $m \leq 2$ at the shipped basis, where the transverse air modes have shapes to overlap against.

2.2. The modal basis

Each head is a lossy tensioned circular membrane with a small bending stiffness, fixed at the rim:

$$\sigma w_{tt} + d_0 w_t - d_2 \Delta w_t + D \Delta^2 w - T \Delta w = f \tag{1}$$

with surface density σ, tension T, bending stiffness D, and the loss coefficients of Equation 19. On a circle of radius R the eigenfunctions are Fourier–Bessel,

$$\varphi_{mn}(r, \theta) = J_m(\alpha_{mn} r / R) \cdot \begin{cases} \cos m\theta \\ \sin m\theta \end{cases} \tag{2}$$

where α_{mn} is the n-th positive zero of J_m, so that the fixed-edge condition $J_m(kR) = 0$ holds by construction. Substituting Equation 2 into Equation 1 gives the dispersion relation the whole instrument's pitch follows from:

$$\omega_{mn}^2 = \frac{T}{\sigma} k_{mn}^2 + \frac{D}{\sigma} k_{mn}^4, \quad k_{mn} = \alpha_{mn} / R \tag{3}$$

This is the reason a pre-solve is possible at all. Equation 3 is closed form: a bank's mode frequencies are read off tension, density and radius without rendering a sample, which is what Section 10 exploits at roughly a hundredth of the cost of one evaluation. Zeros of J_m are found once per process by a sign-change scan and bisection and then memoised, orders 0 to 28 in both indices, because mode generation is otherwise the dominant cost of a fit.

The basis is the membrane's, and the bending stiffness rides on it as a perturbation. A fourth-order operator requires two boundary conditions and its eigenfunctions are combinations of J_m and I_m; pure Fourier–Bessel shapes are the $D = 0$ ones, which is why a single fixed-edge condition can determine them at all. So Equation 3 carries D and Equation 2 does not: the frequencies know about the stiffness, the mode shapes — and with them the modal masses, the strike weights and the radiation weights of Equation 22 — do not. What is neglected is of size Dk^2/T, which at the shipped batter values ($D = 0.001$ N· m, $T = 1250$ N/m) is 0.0149 at the highest retained mode, $k = 136.4$ m^{-1}. That is 0.74% in frequency there, about 12.8 cents — larger than the 0.4% and 0.3% asymmetry splits of Equation 4, which the model treats as audible. Removing the approximation means solving the genuinely clamped fourth-order problem — zero displacement **and** zero slope at the rim — for its own eigenfunctions, and giving up the closed form the pre-solve of Section 10 rests on.

Every $m > 0$ eigenvalue is doubly degenerate on an ideal circle, and no real head is one. A single reduced parameter splits each pair deterministically about its own midpoint,

$$f^{\cos} = f(1 - s/2), \quad f^{\sin} = f(1 + s/2) \tag{4}$$

with the pair's mode shapes evaluated at $\theta - \theta_a$ about a principal tension axis [1]. The shipped splits are 0.4% and 0.3%; zero is an exact compatibility mode, which matters because it is what an older saved configuration decodes to.

The citation carries the phenomenon and not the pattern. Worland measures degenerate pairs **rotated and separated** under non-uniform rim tension, and a lug pattern is not a uniform perturbation: an N-lug drum splits the azimuthal orders that are multiples of N strongly and leaves the others nearly degenerate. One scalar applied to every pair alike cannot produce that. What Equation 4 reproduces is that pairs split, not which ones do.

Selection retains a fixed count, not a fixed bandwidth. Candidates are ordered by frequency and admitted until a slot budget is full, an $m > 0$ mode costing two slots for its two orientations. Since $\omega \propto \sqrt{T}$, rendered bandwidth is a function of tier **and tuning** together — Table 1 is the shipped tuning, and Table 8 is the same budget two octaves down, where the ceiling falls below the band the fit cares about.

Tier	Batter	Resonant	Top mode	Real time
Draft	48	20	929 Hz	—
Standard — *shipped*	96	24	1310 Hz	0.70×
High	160	24	1662 Hz	—

Table 1: What each budget buys at the shipped tuning: the two heads' oscillator counts, the highest mode the head that radiates retains, and the worst-case real-time factor on `js/wasm`. Doubling the batter count buys 0.5 of an octave, which is Equation 24 seen from the other side. The resonant head no longer scales with the tier — it has its own budget, sized against the cavity rather than against a quality setting — so at Draft its own ceiling, 1001 Hz, is the higher of the two. The one measured real-time figure is the Standard worst case with the coupling table at its shipped size; the same configuration measures 1.40× with the coupling off and High 1.15 ×, and Section 2.10 is where that is discussed rather than tabulated.

The resonant head costs a couple of dozen oscillators rather than a second full bank, and the **selection** part of that reduction is exact rather than approximate. Nothing can excite a resonant mode the air cannot reach: the strike reaches only batter modes, and the cavity — the sole path between the heads — couples through an overlap integral whose azimuthal factor is identically zero unless the head mode's order matches a cavity mode's. Their displacement, their contribution to the tension law and their stored energy are therefore zero for all time, so dropping them changes the output not approximately but not at all, and a regression test asserts bit-identical renders rather than a tolerance, at both ends of the cavity basis. The filter is applied **after** selection, not during it: skipping those candidates inside the loop would free their slots and admit higher reachable modes instead, which is a different instrument rather than a cheaper one.

What the reduction retains changed when the cavity did, and the exactness did not. With one lumped pressure state the reachable set is $m = 0$ and the filter is literally "axisymmetric only", leaving 6 of Standard's 96 slots. With the shipped six-state cavity it is $\{0, 1, 2\}$ and the same selection leaves 28. The statement is therefore "the orders the air can reach" rather than "axisymmetric", and it is bit-exact at both ends for the same reason — the selection rule zeroes the coupling of every unreachable order, and those modes provably never leave zero.

The second budget is not exact, and is stated separately for that reason. A head that is only ever driven through the air has no business being sized by a quality tier chosen for the head the stick hits, and sharing one number was accidental — harmless while the reduction left half a dozen modes, and not harmless once a cavity setting started quadrupling the resonant bank. The resonant head therefore has its own limit, 24, and that truncation from 28 **is** a change to the instrument. It is a small one: broadband level moves by less than 0.001 dB and the small-signal transfer function by 0.019 dB RMS to 4 kHz. The value is chosen against the mechanism rather than against a clock — 24 is the smallest bank whose modes straddle both transverse cavity resonances, with the $(1, 1)$ air mode at 660 Hz between resonant partials at 472 and 685 Hz and the $(2, 1)$ at 1094 Hz between 1001 and 1213 Hz, the 23rd and 24th oscillators. Below a budget of 12 the first straddle is missing and the coupling feature of Figure 1 reads 3 dB wrong.

2.3. Coupling through the cavity

The air enclosed by the shell is expanded in the rigid-walled cylinder's own modes. A rigid wall carries no normal velocity, so the radial condition is **Neumann** — and not the Dirichlet condition the clamped heads obey, which is a distinction worth stating because the two families look alike and are not:

$$\Psi_{mn}(r, \theta) = J_m(j'_{mn} r/a) \cdot \begin{cases} \cos m\theta \\ \sin m\theta \end{cases}, \quad \omega_{mn} = c j'_{mn}/a \tag{5}$$

where j'_{mn} is the n-th zero of $J_{m'}$. The uniform mode is the $m = 0$ member at $j' = 0$: a mode of zero frequency, and the whole of the lumped spring this replaced. Note that $j'_{01} = 3.8317$, the first **positive** zero of $J_{0'} = -J_1$, is a different and non-degenerate mode.

Axial order is excluded deliberately, not overlooked. A pressure varying along the shell would make the two heads see different pressures — the coupling coefficient picks up a factor that is $+1$ at the batter head and $(-1)^l$ at the resonant one instead of the same at both, which is a larger change than this one — and the first axial mode sits at $c/2L = 858$ Hz on the shipped 0.2 m depth, above the transverse modes this exists to add. Axially uniform is a first cut and is described as one.

Cavity mode	j'	at 0.1524 m	at 0.1584 m
$(0, 0)$ uniform	0	0 Hz	0 Hz
$(1, 1)$ cos/sin	1.8412	659.5 Hz	634.5 Hz
$(2, 1)$ cos/sin	3.0542	1094.0 Hz	1052.6 Hz
$(0, 1)$	3.8317	1372.5 Hz	1320.5 Hz

Table 2: The six shipped air states, at the shipped 12-inch radius and at the 0.1584 m the excitation-gap analysis states its hypothesis for. The second column is where that analysis compares the reference's 624.4, 1018.4 and 1331.3 Hz partials; a test pins the generated table to it. The series is $cj'/2\pi a$ and nothing else, so it is set by the shell and the air and not by the tuning.

Each head mode couples to each air mode through the overlap of their shapes over the head's disc, and the same coefficient appears in both directions — the air is driven by $C_{ic}\dot{q}_i$ and the head is loaded by $C_{ic}P_c$:

$$C_{ic} = \int_A \varphi_i \Psi_c \, \mathrm{d}A \tag{6}$$

The angular integral separates and gives a **selection rule**: the coefficient vanishes unless the azimuthal orders match, and at an unrotated principal tension axis unless the orientations match too, a rotated axis entering as a plane rotation through $m\psi$ and therefore as an isometry that leaves the total coupling strength alone. A head mode reaches at most one air mode per radial order, which is what makes the extension affordable: 44 of the 768 head/air pairs at the shipped basis are non-zero. Against the uniform mode $\Psi = 1$ and Equation 6 collapses to the signed swept area,

$$A_{0n} = 2\pi R^2 J_1(\alpha_{0n})/\alpha_{0n}, \quad A_{mn} = 0 \ \text{ for } \ m > 0 \tag{7}$$

which is the closed form the lumped model always used and which the code returns verbatim in that case. For every other air mode the radial integral has no clean closed form — the two Bessel functions carry different arguments and different boundary conditions, so the Lommel collapse does not happen — and is evaluated by 96-point Gauss–Legendre quadrature once per coupled pair at construction. Nothing in the render touches it, and a test checks the quadrature against the analytic swept area where the two must agree, to 2×10^{-16}.

The coupled system is a diagonal modal bank plus a (P_c, H_c) pair per air mode:

$$\ddot{q}_i + 2\gamma_i \dot{q}_i + \omega_i^2 q_i = f_i/M_i - \sum_c C_{ic} P_c / M_i \tag{8}$$

$$\dot{P}_c = K_c \sum_i C_{ic} \dot{q}_i + \omega_c H_c - \lambda P_c, \quad \dot{H}_c = -\omega_c P_c \tag{9}$$

$$K_c = s\rho c^2/\Lambda_c, \quad \Lambda_c = \int_V \Psi_c^2 \, \mathrm{d}V \tag{10}$$

with $\Lambda_{00} = V = \pi R^2 L$ the shell volume and λ a pressure loss. Writing each mode in that first-order pair rather than as a displacement/velocity one is what makes the uniform member degenerate cleanly: $\omega_{00} = 0$, so H never leaves zero and Equation 9 collapses to the single $\dot{p} + \lambda p = K \sum_i A_i \dot{q}_i$ the lumped model had. The system is passive by construction: with

$$E = \sum_i \frac{M_i}{2}(\dot{q}_i^2 + \omega_i^2 q_i^2) + \sum_c \frac{P_c^2 + H_c^2}{2K_c} \tag{11}$$

one has $\dot{E} = -\sum_i 2\gamma_i M_i \dot{q}_i^2 - \lambda \sum_c P_c^2/K_c \le 0$. That is **proved rather than assumed**, and the proof is the reason Equation 6 is written with one symbol: the head equation loses $\sum_c P_c \sum_i C_{ic}\dot{q}_i$ and the cavity equation gains exactly that quantity, because the same coefficient stands in the drive and in the load, so the coupling terms cancel identically for any number of air modes and any s. The ω_c rotation cancels against itself for the same reason. The air can exchange energy between the heads but never manufacture it.

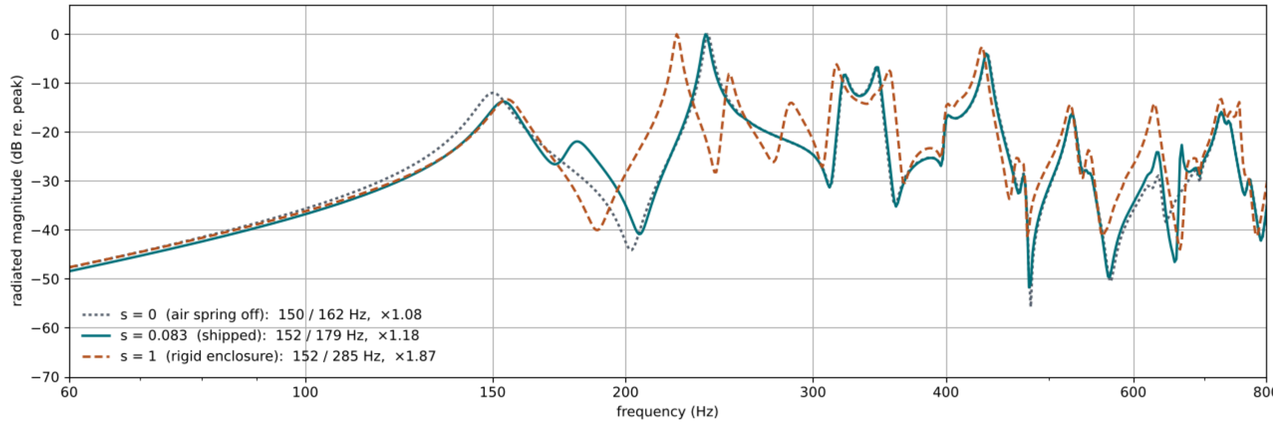

Figure 1: The continuous-time radiated response at three cavity stiffnesses. The doublet opens with s while its lower member stays penned between the two heads' own fundamentals, which is eigenvalue interlacing and is asserted by test. Above the axisymmetric family the curves no longer coincide, and that is the transverse air modes: at the $(1,1)$ of Table 2 the shipped cavity moves the response 12.3 dB against no air spring at all, where the lumped cavity moved it 0.04 dB. Equation 7 is now the statement about **one** member of the expansion rather than about the cavity.

The stiffness scale s is fitted, and the fact that it has to be is the interesting part. The rigid formula $\rho c^2/V$ assumes a sealed, rigid shell driven by two pistons. A real shell flexes and the heads are not pistons, and the formula correspondingly over-predicts the stiffening badly: at $s = 1$ the axisymmetric fundamental splits into branches 1.87 apart, where measured two-headed drums separate their two $(0,1)$ branches by 10–20% [2] — Fischer measured 186 Hz with one head and 215 Hz once the resonant head was added at unchanged tuning, a ratio of 1.16 [3]. The shipped $s = 0.083$ gives 1.18. Figure 1 is the whole effect.

A vent was on the list of excuses for s, and it does not belong there. A vent is a Helmholtz port and therefore a **high-pass** leak: below f_H the air flows out and no pressure builds, above it the plug's inertia blocks the flow and the cavity behaves as sealed. On this shell f_H runs 21.8 Hz for a 6 mm vent to 59.6 Hz for a 25 mm one, all far below the 150 Hz fundamental, so the diverted fraction there — falling as $1/f^2$ above f_H — is 2.1% to 15.8%, and less at every higher mode. Explaining a twelvefold reduction needs about 92%. The honest list is therefore shell flex and the non-piston mode shape, with the vent worth a few per cent at most; neither of the two has been measured separately here, and that is the open question the fitted s now stands for.

s is a fraction rather than a free gain because the rigid, sealed, piston-driven enclosure is the stiffest case that exists: 1 is a physical ceiling, not a neutral setting, and a fitted value above it would be a statement that the model is wrong rather than that the drum is stiff. It multiplies every K_c alike, so the ceiling keeps its meaning once the air carries more than one state.

The most useful result of widening the cavity is a negative one. The hypothesis the work was done on was that the lumped reduction mis-set the compliance, and that this was why the fitted s sits a factor of twelve below the rigid ceiling: one uniform pressure state stands in for a whole field, and there is no obvious reason its best-fit stiffness should be the true bulk value. It is. Every non-uniform air mode has exactly zero net volume — $\int_A \Psi_c \, \mathrm{d}A = 0$ for $j' \neq 0$ — so its impedance $K_c(j\Omega)^2/((j\Omega)^2 + \lambda j\Omega + \omega_c^2)$ vanishes as $\Omega \to 0$ and it contributes nothing to the quasi-static air spring the doublet measures. The static compliance of the full expansion is **exactly** $\rho c^2/V$, and it always was. Measured rather than argued: the doublet is 155.3 / 178.7 Hz, a ratio of 1.1509, and it is 1.1509 with one air state and with the shipped six, unmoved to the transform's 2.93 Hz resolution. s would still have to be 0.083. The transverse modes are resonators, not springs; they reshape the response near their own frequencies and leave the quantity the fit is made against alone.

2.4. The tension nonlinearity

Striking a drum harder stretches its head, which raises its tension, which raises every mode frequency together — and the pitch falls back as the hit decays. That downward glide is the characteristic feature of a tom, and it is modelled by a Berger-style reduced tension law over the modal strain measure

$$S = \int |\nabla w|^2 \, \mathrm{d}A = \sum_i \Gamma_i q_i^2, \quad \Gamma_i = M_i k_i^2/\sigma \tag{12}$$

$$\begin{aligned} \Delta T(S) &= T_{\max} \tanh(\beta S/T_{\max}) \\ U(S) &= \frac{T_{\max}^2}{2\beta} \log\cosh(\beta S/T_{\max}) \end{aligned} \tag{13}$$

Near rest this is the ordinary Berger law $\Delta T = \beta S$ with $U = \beta S^2/4$ — the reduction family in which the short-time tension variation is proportional to the system's energy [4].

The approximation here is worth locating precisely, because it is not in Equation 12. That diagonal form is exact: the mode shapes are Dirichlet Laplacian eigenfunctions, so $\int \nabla\varphi_i \cdot \nabla\varphi_j \, \mathrm{d}A = k_i^2 \int \varphi_i\varphi_j \, \mathrm{d}A$ and the cross terms vanish identically. Writing $g = |\nabla w|^2$, all of Berger's error lives in the **second** moment: the quartic membrane potential goes as $\int g^2 \, \mathrm{d}A$, while Equation 13 uses $\left(\int g \, \mathrm{d}A\right)^2/A$, the projection of g onto the constant function. Being a projection rather than a truncated series, it satisfies $U_{\text{Berger}} \leq U_{\text{exact}}$ by Cauchy–Schwarz. Section 2.5 is that second moment put back.

The tanh is a smooth cap rather than a clip, so it bounds the frequency excursion without discarding stored energy. Each mode is detuned by $\Delta\omega_i^2 = \Delta T k_i^2/\sigma$.

The cap earns its keep at the anti-alias bound. Frequencies scale as $\sqrt{1+r}$ where $r = T_{\max}/T_0$, so retaining modes up to a fraction ν of the sample rate requires

$$r < 1/(4\nu^2) - 1 \tag{14}$$

which at the shipped $\nu = 0.45$ is 0.2346 against a shipped r of 0.2 — a 17% margin, validated at configuration decode rather than assumed. Equation 14 bounds a **uniform** detune and nothing else, which is why the coupling of Section 2.5 needs its own bound. At the shipped coefficients the glide is 104.9 cents on the loudest hit and 3.0 on the quietest, an audible semitone that still leaves the plateau clear: past roughly twice these coefficients a loud hit sits on the flat of the tanh, which turns the glide into a hold-then-drop and erodes the velocity dependence that makes it expressive.

Because the tension depends on the state at the end of the step and the state depends on the tension, the update is implicit. It is closed with a **discrete gradient** rather than by evaluating Equation 13 at either endpoint:

$$\overline{\Delta T} = 2[U(S^{n+1}) - U(S^n)]/(S^{n+1} - S^n) \tag{15}$$

Paired with the midpoint displacement this makes the nonlinear work over a step exactly equal the change in stored potential, so the lossless model conserves Equation 11 to solver tolerance rather than drifting — which is what lets an energy assertion be a test rather than a hope.

2.5. The coupling channels

Equation 13 is **exactly** the rank-one projection of the quartic potential onto the constant channel. That is a stronger statement than "an approximation", and it is what makes the missing part addable rather than replaceable. Choose an orthonormal set ψ_c on the head and write

$$\hat{g}_c = \langle g, \psi_c \rangle = q^\top D^c q, \quad D^c_{ij} = \int \psi_c \big(\nabla\varphi_i \cdot \nabla\varphi_j\big)\, \mathrm{d}A \tag{16}$$

$$U = \frac{\tilde{\beta}}{4} \sum_c \hat{g}_c^2 \tag{17}$$

The uniform channel $\psi_0 = 1/\sqrt{A}$ gives $D^0 = \mathrm{diag}(\Gamma_i)/\sqrt{A}$ and $\hat{g}_0 = S/\sqrt{A}$, so its term **is** the Berger potential with $\beta = \tilde{\beta}/A$ — not an approximation to it, because the mode gradients are orthogonal analytically by Green's identity. The table therefore stores only $c \geq 1$ and adds to the shipped capped law rather than replacing it, which is also why the tanh stays exactly where it was needed, on the channel that detunes every mode uniformly, and is not applied to channels that detune nothing uniformly.

The force is cubic and odd, and that fixes what it can reach. An even potential gives an odd force, so the combinations generated are $3f_a$, $2f_a \pm f_b$ and $f_a \pm f_b \pm f_c$ — no second harmonic and no simple sum or difference tone, which would need a **quadratic** term in the potential that a shell or a curved plate has and a flat tensioned head does not. The consequence is a design requirement rather than a preference: the lowest combination consumes three slots, so a single pump reaches only f_a and $3f_a$, and $3f_{01} \approx 450$ Hz falls **below** the 476–700 Hz band the excitation gap of Section 11 sits in. At least two pumps are therefore mandatory, and a configuration asking for fewer is rejected outright with that reason.

The channels are built from $\{1\} \cup \{\nabla\varphi_a \cdot \nabla\varphi_b : a, b \in P\}$, Gram–Schmidt orthonormalised offline, with each D^c truncated to entries carrying at least one index in the pump set P — so the $|P| \geq 2$ requirement is structural rather than documented. The selection rule falls out of the angular algebra: the gradient product of two Fourier–Bessel modes is exactly two harmonics, at $|m_a - m_b|$ and $m_a + m_b$, so a quartic coefficient vanishes unless two gradient products share an angular order **and** an orientation family. That second condition is the one that removes most of the tensor, it is **not** the naive $\pm m_i \pm m_j \pm m_k \pm m_l = 0$ the four-index form suggests, and there is no radial rule at all. On the shipped bank with $|P| = 4$ it leaves 408 structurally non-zero coefficients across 10 channels, removing about 89% of the candidates, and 34 modes provably carry none. The shipped budget retains 256 of the 408.

The pump set is chosen by displacement, not by frequency and not by energy. Since the force goes as q^3, the right weight is peak modal displacement under a reference velocity-1 strike, available in closed form from the strike weight and the contact pulse's transform. The measured ranking is not frequency-ordered and that matters: the $(2, 2)$ at 525 Hz outranks the $(1, 1)$ sine at 240 Hz, which outranks the $(1, 2)$ at 437 Hz, so a frequency-ordered set of four would retain different modes. The shipped four are the $(0, 1)$ at 150.1 Hz, the $(1, 1)$ cosine at 238.7, the $(2, 1)$ cosine at 320.0 and the $(0, 2)$ at 344.7.

The energy exactness survives the extension, and for free. Because U is a sum of functions of **scalar** quadratic forms, the scalar secant of Equation 15 already is the vector discrete gradient, applied per channel: no Gonzalez projection, and no $0/0$ branch to take on a 96-vector at rest. Measured, the identity holds to a relative residual of 2.5×10^{-15} and the lossless coupled system drifts by 1.1×10^{-11} over a second.

The anti-alias bound has to be restated, because a force that multiplies three sampled signals reaches the sum of three modal frequencies and Equation 14 says nothing about that. Since every retained entry carries a pump index the worst case is not $3f_{\mathrm{top}}$; a receiver that is itself a pump admits two free indices and is bounded by

$$f_{\max P} + 2f_{\mathrm{top}} \tag{18}$$

while a free receiver reaches only $2f_{\max P} + f_{\mathrm{top}}$. At the shipped bank the conservative bound is 3631 Hz and the table actually built reaches 2709 Hz, both against 21.6 kHz — an eightfold margin, not binding at any shipped tier. It is implemented anyway, because it bites if the tier rises, if the sample rate falls toward 8 kHz, or if the pump band widens.

What it does is measured on modal amplitudes rather than on the spectrum, and the distinction is the finding. With every non-pump mode's strike projection zeroed, so that the cubic force is the **only** path into any other mode, modes that receive no excitation at all rise by roughly 50 dB: the $(2, 2)$ cosine at 524.9 Hz from -76.3 to -28.4 dB, and the $(2, 3)$ cosine at 725.4 Hz from -86.4 to -34.6. Reading that off the radiated spectrum would have flattered it, because four damped sinusoids put a Lorentzian leakage floor at -37 dB across the band — leakage from the pumps rather than content in it. With the cavity disabled the uncoupled figures are not small but **exactly zero**: those modes never move.

That is the mechanism Section 11 leaves open. A mode pumped by the coupling does not read $|F(f)|$ at its own frequency at all, so it can be excited precisely where the half-sine's zero comb has deleted the excitation outright — which is the one thing the excitation-gap analysis never considered, because the model had none to consider.

One independent corroboration, and one calibration that is now pending. At the shipped configuration $\tilde{\beta} = \beta A = 7.0 \times 10^5$ N/m, against a material $Eh/\big(2(1-\nu^2)\big)$ of 6.5×10^5 N/m for a mylar head of the implied thickness — agreement to about 8%, which is the strongest independent corroboration any coefficient in this model has. It is worth two caveats: it reuses the same unmeasured surface density and a datasheet modulus, so it is not fully independent, and the coefficient is fitted either way. Against that, the glide moved only 102.8 to 104.9 cents, far less than the ratio between the two moments predicts, and the reason is the fixture rather than the model: the calibration hit is at the head's **centre**, where only axisymmetric modes are excited and the orthogonal channels see almost nothing. The shipped strike is off-centre, so the calibration is recorded as pending a refit rather than as confirmed.

And the local quartic is a bracket, not the exact operator. Full von Kármán condenses the in-plane displacement through an Airy stress function, giving a quartic with an inverse-biharmonic kernel. Berger is that family's **uniform** limit, with the in-plane stress spatially constant, and $\int g^2\, \mathrm{d}A$ its **local** limit, with the stress following the local slope and no elastic redistribution at all; the truth sits between. What is claimed here is therefore the coupling's **structure** — which frequencies can be generated, and by which mode sets — and not its magnitude.

2.6. The loss law

Each mode's decay rate is the sum of three separately meaningful channels plus an optional measured residual:

$$\gamma_i = \underbrace{d_0 + d_1 k_i + d_2 k_i^2}_{\text{structural}} + d_{\mathrm{rad}} \alpha_i^2 + \Delta_i \tag{19}$$

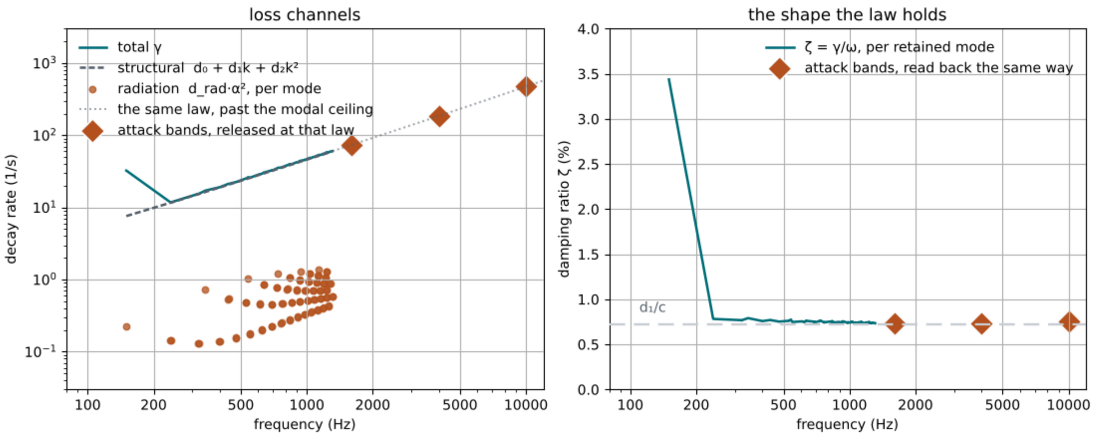

Figure 2: Left: the loss channels of the batter head, with the structural law continued past the highest resolved mode and the attack layer's three bands sitting on that continuation. Right: the same law as the quantity it is shaped to hold constant. The $(0, 1)$ is the one mode that is deliberately off the law — see the text — and everything else lies within a few hundredths of a per cent of d_1/c, including three bands that are not modes at all.

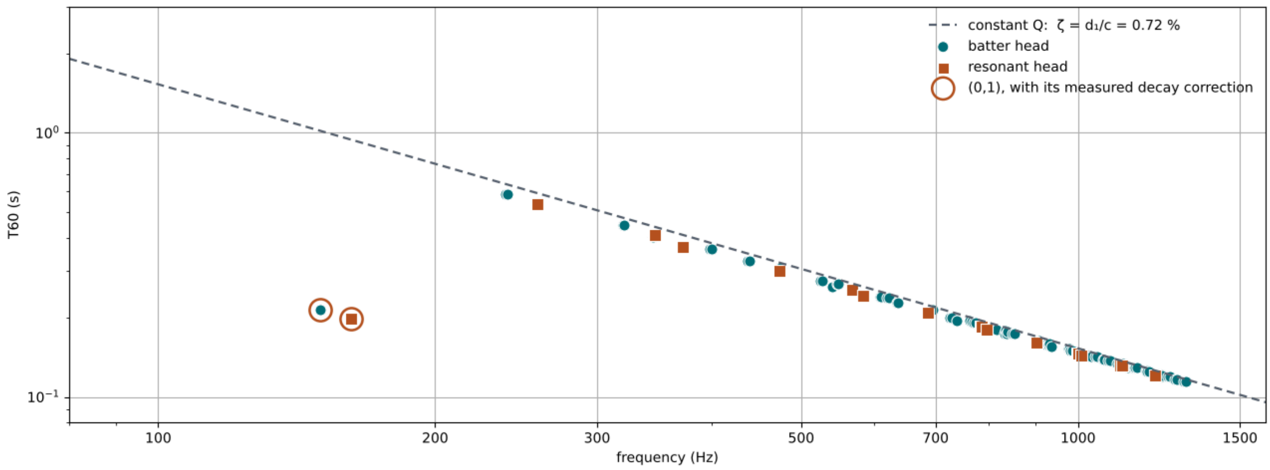

Figure 3: The whole instrument's mode map at the shipped tuning: 96 batter and 24 resonant oscillators, against the constant-Q law implied by d_1/c alone — a line derived from two coefficients, not fitted to the markers it passes through. This is the model-side counterpart of Figure 9 and is drawn on the same axes.

from which $T_{60} = \ln 1000/\gamma_i$. They are kept apart rather than collapsed so that calibration can attribute a change to a cause, and the k^1 term is the one that carries the physics. With $\omega \approx ck$ on a membrane, the fraction of critical damping is

$$\zeta_i = \frac{\gamma_i}{\omega_i} \approx \frac{d_0}{ck} + \frac{d_1}{c} + d_2\frac{k}{c} \tag{20}$$

so d_1 alone expresses **constant Q**: d_0 alone gives a T_{60} independent of frequency and d_2 alone gives $T_{60} \propto 1/f^2$, where the measured membrane behaviour is $T_{60} \propto 1/f$ [5]. The shipped coefficients set $\zeta = d_1/c = 0.72\%$, and across the whole retained band the realised ζ stays between 0.73% and 0.80% (Figure 2). d_0 and d_2 are deliberately small: each flattens or steepens the law at one end, and both are corrections to a shape the k^1 term already has right.

One mode is deliberately exempt. A two-headed drum loses its axisymmetric fundamental fastest of all, into the cavity and the opposite head, and the reduced cavity does not reproduce the full magnitude of that path — and widening it to six states does not either, since every state but the uniform one has zero net volume and so carries nothing at this frequency. The residual is carried explicitly, as a per-mode correction Δ_i on the $(0, 1)$ of each head, fitted to hold its T_{60} at 0.213 s. It shows in Equation 20 as a ζ of 3.44% against the band's 0.74%, and in Figure 3 as the one marker far below the line. Naming it as a correction rather than folding it into d_0 is what keeps the other two coefficients readable.

Two consequences reach the rest of the paper. First, tuning must not be a sustain control: because $c = \sqrt{T/\sigma}$ appears in Equation 20, holding d_1 fixed while the tuning knob moves T would drag ζ from 2.2% to 0.7% across the knob's travel and stretch a 300 Hz partial's T_{60} from 0.166 s to 0.423 s. Retuning therefore rescales d_1, d_2 and every correction by $\sqrt{T_{\text{new}}/T_{\text{old}}}$, so the knob moves pitch and T_{60} falls only in the proportion constant Q requires.

Second, and this is what Section 9 turns on: the product's damping control scales this whole law, and a decay tilt slopes it. A residual that is neither — a curve that falls and rises again, which is what the reference's non-monotone ring times demand — is a statement about the **form** of Equation 19, and the place an answer would go is the per-mode correction term that already exists for exactly one mode.

2.7. The strike

Excitation is selectable, and the two options differ in what they take as given. The shipped `prescribed` model writes a half-sine of a measured contact duration into the force buffer at trigger time, impulse-normalised in closed form, with the duration running 8

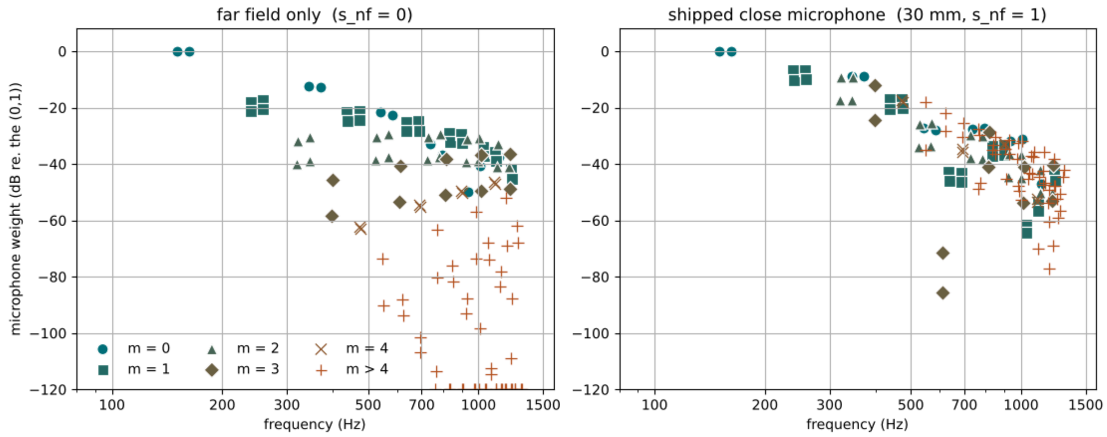

Figure 4: Partial balance by azimuthal order, with and without the near-field term, each normalised to its own $(0,1)$. In the far field a 12-inch head below its modal ceiling is very nearly a monopole: the $(1,1)$ pair sits 18 and 21 dB down and the higher orders cascade into the numerical floor. At 30 mm the evanescent term restores them — $(1,1)$ at -6.8 and -10.1 dB, $(0,2)$ at -8.7 — which is what a tom microphone actually hears. Both heads' modes are plotted, so the second $m=0$ marker near 162 Hz and the $m \leq 2$ resonant series are weights the resonant head **would** radiate with and which this model deliberately does not sum in.

ms at the quietest hit to 5.5 ms at the loudest and scaled by stick hardness. The head never influences it.

The `hertzian` model integrates the stick as a free mass against a Hunt–Crossley contact [6], [7],

$$F = K[\delta]_+^{\alpha}\left(1 + h\dot{\delta}\right), \quad \delta = z - w \tag{21}$$

where the elastic force is scaled by a factor linear in compression **rate**, so the loss vanishes with the compression instead of stepping at impact and at separation the way a linear dashpot does. Duration, pulse shape and re-contact become outputs. The exponent is **measured rather than assumed**: Hertzian contact time scales as $v^{-(\alpha-1)/(\alpha+1)}$, and Wagner's crescendo runs 7.5 ms at piano to 5.9 ms at forte over a three- to fourfold velocity range [8], which implies α between 1.42 and 1.56. The canonical spherical $3/2$ is therefore what the velocity dependence says, not a convenience — which also means the prescribed model's velocity law is reproduced by this change rather than discarded by it.

The contact time is set by the head, not by the tip. The driving-point mass of the batter head, $1/\sum_i a_i^2 M_i$, is 0.31 g against a 15 g mallet, and the same stick that rebounds off a rigid target in 0.40 ms stays in contact with the head for 7.26 ms. Two decades of contact stiffness span a factor of 1.5 in duration and then stop mattering, because the stick is riding the head's own return rather than rebounding off its own compression. It is the same fact as a stick being pressable into a roll.

What the principled model bought is measured, and it is not the low band. A prescribed half-sine of duration τ has magnitude spectrum $|\cos(\pi f\tau)|/\left|1-(2f\tau)^2\right|$, which is a comb of **exact zeros** at every $(k+1/2)/\tau$ rather than a tilt: at the fitted τ of 8.23 ms, 547 and 668 Hz sit at -309 and -315 dB, which is why nothing downstream can lift them. Hertzian contact shallows that comb and moves it but does not remove it, because it is still one smooth touch — measured, it is worth 0–4 dB below 700 Hz and 12–23 dB above 800 Hz.

Part of that advantage has since been taken from the other end. With the coupling of Section 2.5 active the Hertzian model's lead at 800 Hz falls from 11.9 to 7.9 dB, with 1500 and 2500 Hz unmoved, and it falls because the **prescribed** side rose about 4 dB rather than because the Hertzian side lost anything. The cubic force fills part of the band the half-sine's zero comb deleted, which is the excitation gap being addressed by a source instead of by a touch.

Neither model reproduces what the measured interval actually contains: Wagner's 5.5–8 ms is a **dwell** time spanning three separate impacts — the strike, the wave it launched returning off the rim about 1.7 ms later, and the re-loaded contact [8] — and both models spend that interval on a single force pulse. The Hertzian version does not produce those re-contacts either; a build that appeared to was a discretisation artifact and it converged away.

Whichever model is selected, the force is injected through the mode shapes times a finite footprint factor $2J_1(ka_c)/(ka_c)$, and the strike-point state is read back through the **same** weights. That reciprocity is not decoration: it makes $F \cdot v$ exactly the power the modes receive, so contact can neither create nor destroy energy.

2.8. What the microphone hears

The observable is **volume acceleration**. Far-field pressure from a compact source is proportional to it with no further frequency dependence, so the weight applied to each mode's discrete acceleration is a pure geometric factor. For a circular mode that factor is a Lommel integral, and because $J_m(\alpha_{mn}) = 0$ by construction it collapses with no series left over:

$$G_{mn} = 2\pi R^2 \alpha_{mn} J_{m+1}(\alpha_{mn}) J_m \frac{u}{\alpha_{mn}^2 - u^2} \tag{22}$$

with $u = \omega_{mn} R \sin\theta / c_{\text{air}}$ the acoustic trace wavenumber across the head. Two limits carry the physics. At $u=0$ this is **exactly** the swept area of Equation 7 for $m=0$ and exactly zero for $m>0$ — an on-axis microphone hears the axisymmetric series and nothing else. And $J_m(u) \sim (u/2)^m/m!$ for small u, so multipole cancellation falls out of the integral rather than being approximated by a fitted roll-off exponent.

That distinction is a near-miss worth recording. An earlier version of this model summed modal **velocity** weighted by a radiation efficiency $\left(ka/\sqrt{1+(ka)^2}\right)^{m+1}$. As an amplitude ratio against velocity, that efficiency already contains one factor of ka, so reusing it beside a volume-acceleration observable counts that factor twice; and raising it to $m+1$ stands in for a cancellation whose true magnitude is $1/(2^m m!)$, which at the highest retained azimuthal order is wrong by seven orders. A fitted output gain would have absorbed the level error and hidden both. The efficiency survives, but only where it belongs — apportioning radiation damping across the series in Equation 19.

The total weight is a **sum** of two mechanisms, not a product:

$$W_i = \frac{G_{mn} D_m}{1 + d/R} + s_{\text{nf}} \cdot 2\pi R^2 e^{-\alpha_{mn} d/R} \Phi_i \tag{23}$$

Here D_m is the azimuthal directivity at the microphone's angle, d its distance and Φ_i the mode shape beneath it. The second term is the non-propagating near field, and a close-miked tom is mostly made of it. Figure 4 is why it has to exist.

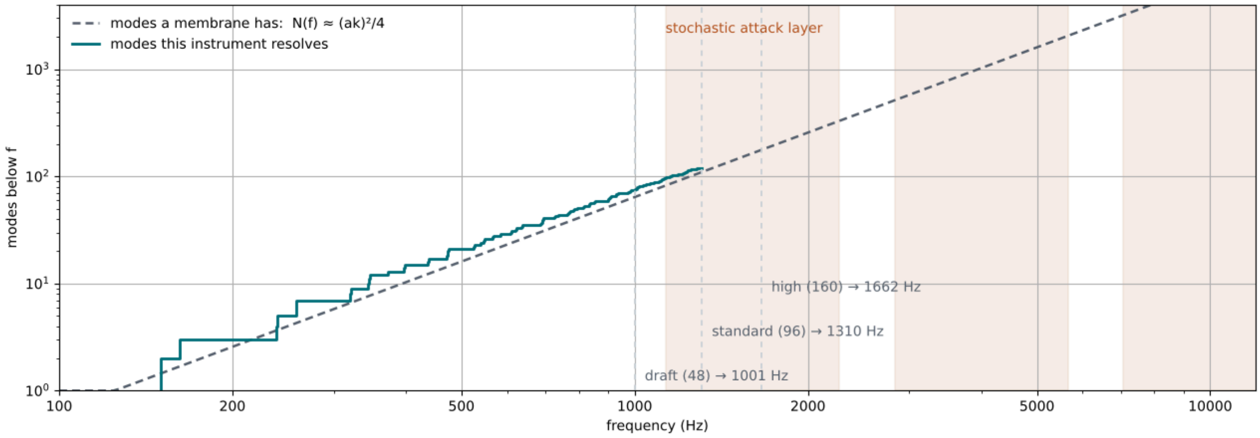

Figure 5: Why the instrument is a hybrid. The staircase is what the budget resolves across both heads, the dashed curve is Equation 24 for one membrane — what would have to be resolved to keep going — and the shaded spans are what the stochastic layer covers instead. The two track each other in slope up to the ceiling, which is the check that the count is being spent on the modes a membrane actually has; the staircase's excess above the curve is the second head's 24 oscillators.

s_{nf} is fitted, not derived: the effective area of an evanescent patch is outside what a reduced model can compute. It matters more than any other single number here, and it is also why the fit of Section 8 "shifting the pickup" is a substantive move rather than a cosmetic one — microphone radius and angle enter both terms of Equation 23, so they trade the entire partial balance, not a level. A Butterworth high-pass at 35 Hz and low-pass at 12 kHz and a scalar gain complete the chain.

2.9. The attack layer

Modal synthesis cannot reach a drum's real bandwidth in a browser, and the reason is a counting argument rather than an engineering one. A membrane's mode count below a frequency grows as its square,

$$N(f) \approx (Rk)^2/4, \quad k = 2\pi f/c \tag{24}$$

so this head needs about 130 oscillators to resolve 1 kHz, 530 for 2 kHz and 3300 for 5 kHz, against a budget of 96 (Figure 5). Equivalently: bandwidth grows only as the square root of the count, so doubling the oscillators buys about half an octave — which is Table 1 measured rather than derived.

What lives above the ceiling is not individually resolvable anyway: it is a dense, fast-decaying thicket that reads as the stick rather than as pitch, which is precisely why the published tom analysis finds five to ten modes sufficient for the **sustain** of a central strike [9]. It is therefore modelled as band-limited noise driven by the same contact force that drives the modes — which means hardness and velocity carry into it for free, with no second set of parameters to keep consistent.

Three bands, not one, at 0.4, 1 and 2.5 times a placement frequency. The reason is Equation 19: γ grows with k, so the top of the covered range dies several times faster than the bottom, and a single release makes the whole span ring for as long as its slowest part — heard as a noise burst sitting on the drum rather than as the drum's attack. **The releases are derived, not fitted.** Each band decays at the batter head's own structural law evaluated at that band's centre wavenumber,

$$\tau_b = \frac{\text{scale}}{\gamma_{\text{struct}}}(2\pi f_b/c) \tag{25}$$

so the layer is an extrapolation of the mode series rather than an independent effect beside it, and the damping controls reach it without being routed there. At the shipped placement the three bands land at 1.6, 4 and 10 kHz with T_{60} of 94, 37 and 15 ms — against the flat 138 ms of the single fitted release this replaced, which was about twice too long at the bottom of its range and seven times too long at the top. The right panel of Figure 2 is that claim as a picture: the three bands sit on the same constant-Q line the modes do.

The noise source is a fixed-seed xorshift64*, chosen because it is a few instructions, has one word of state, and is exactly reproducible across platforms — which the language's global source is not, and which the bit-exact render tests require.

2.10. Integration and cost

There are two update paths, and which one runs is decided by the configuration rather than by a quality setting. With the cavity and the nonlinearity both off, the system is a bank of independent linear oscillators and each is advanced by the **exact** matrix exponential of its own second-order system — closed form in three branches by damping regime, with no discretisation error at all.

Otherwise the coupled path runs the implicit midpoint rule. Writing $q^{n+1} = q^n + h v_{\text{mid}}$ and $v^{n+1} = 2v_{\text{mid}} - v^n$ and substituting into Equation 8 gives, per mode,

$$\begin{aligned} & v_{\text{mid}}\left(2/h + 2\gamma_i + \omega_{\text{eff}}^2 h/2\right) \\ & = 2v_i^n/h - \omega_{\text{eff}}^2 q_i^n + f_i/M_i \end{aligned} \tag{26}$$

where ω_{eff} carries mode i's current tension increase. The cavity would couple every mode to every other, but it does so through k rank-one terms, so it is eliminated rather than solved: each mode's midpoint velocity depends on the pressures affinely, and substituting that into the discretised Equation 9 leaves a $k \times k$ system where the lumped model left one scalar equation. That system is $\operatorname{diag}(K_c)$ times a symmetric positive definite matrix — the diagonal term is strictly positive and the coupling block is a Gram matrix with positive weights — so every pivot is positive and elimination without pivoting is safe. k is capped at eight and validated at decode. At $k = 1$ the elimination is literally the old single division, which is what keeps a one-mode cavity bit-exact rather than merely equivalent; a test writes the rank-one form out by hand and compares to the last bit over four thousand samples.

The nonlinearity closes the loop with a fixed-point iteration on Equation 15, capped at eight passes, with the coupling of Section 2.5 inside that loop since its per-channel secants depend on the endpoint too. The cap is not the cost: measured on the shipped configuration at full velocity the mean is 2.88 iterations, and sweeping the tension coefficient over a 32× range moves it only to 3.09, because a stiffer law both perturbs the tension more and contracts faster once the tanh begins to saturate. That measurement retired a planned change — an explicit energy-proportional detune, proposed on the

Quantity	Provenance	What fixes it
Mode frequencies	derived	Equation 3 from T, σ, R
Modal masses, strike weights	derived	orthogonality of Equation 2
Swept areas	derived	Equation 7; exactly zero for $m > 0$
Cavity mode frequencies	derived	Equation 5, zeros of $J_{m'}$ over the shell radius
Head/air coefficients C_{ic}	derived	Equation 6; closed form at $j' = 0$, quadrature elsewhere
Quartic coefficients D^c	derived	Equation 16 over the retained shapes
Far-field weights	derived	Equation 22, a Lommel integral in closed form
Attack-band releases	derived	Equation 25, the head's own Equation 19
Coupling coefficient $\tilde{\beta}$	derived, then corroborated	βA from the fitted Berger β; within 8% of $Eh/(2(1-\nu^2))$
Contact exponent $\alpha = 3/2$	measured	velocity dependence of contact time [8]
Contact duration	measured	5.5–8 ms on a 12-inch tom [10]
Damping ratio ζ	measured	constant Q on membranes [5]
Cavity split ratio	measured	10–20% on two-headed drums [2], [3]
Cavity stiffness scale s	**fitted**	to that measured split
$(0,1)$ decay correction Δ_i	**fitted**	to a 0.213 s ring time
Near-field scale s_{nf}	**fitted**	to partial balance at the shipped mic
Attack level	**fitted**	to spectral balance against the modal layer
Output gain	**fitted**	so a full-velocity hit peaks below clipping
Berger β	**fitted**	to a 100-cent glide, and **pending a refit** — see Section 2.5

Table 3: Where each number comes from. The six fitted rows are the model's admissions: each is a quantity a reduced model cannot compute, and each is fitted against a **different** measurement rather than against the same one. The coupling coefficient is the one row of its own kind — it is not a new free parameter but the fitted Berger one spent on the channels Berger projects away, and it then lands within 8% of a material value it was not fitted to.

assumption that all eight passes ran — and it is the reason the exact energy bookkeeping was kept.

The shipped configuration is below real time in the browser, and that is stated plainly rather than buried. The `js/wasm` worst case — a full-velocity retrigger before every 512-sample chunk, so the solve never idles — measures 0.70× real time at the shipped 256 coefficients, against 1.40× with the coupling off and 1.66× before this chapter's two mechanisms existed. Three things soften it and none of them settle it. It is the worst case and not the steady one, since a real hit lets the solve idle at 3.9×. The cost is not a harder solve: the mean iteration count moves only 2.40 to 2.49 at velocity 1, so what is being paid for is the table walk itself. And nothing here is optimised — the table is walked over a run partition that blocks by channel and by the row of each retained pair, so within a run that mode's displacement and inverse mass hoist out of the inner loop and its force accumulates in a register, but the receiver side is unblocked and still scatters into the acceleration array, and the table is rebuilt per iteration rather than updated incrementally. The budget is also not currently buying much: across a sixfold range of retained coefficients the level in the band the coupling exists to reach moves 0.9 dB, and the full table is not even the loudest of them, so 256 is chosen for margin and 128 is the first thing to try. Making this affordable is open work, deliberately deferred, and not a closed question.

2.11. Derived, measured, and fitted

The distinction the rest of this paper makes between a search question, a range question and a model-structure question needs a matching distinction inside the model. Table 3 is it.

The shipped values themselves are collected in Table 4, Table 5 and Table 6. All of them are SI-valued and versioned: the persisted schema is at version 11, and every earlier version has an explicit migration. Those migrations are not uniformly faithful, and the difference is recorded rather than smoothed — a zero-valued asymmetry or an absent nonlinearity reproduces the old sound exactly, whereas the correction to the microphone model cannot, because what it replaced was not a physical quantity and has no image under the new one.

2.12. What the model is not

Stating the exclusions is part of describing the instrument, because several of them are things a reader would otherwise assume are there.

The resonant head does not radiate into the output. It is fully coupled into the dynamics through Equation 9 and it is audible — as the cavity branch, and as the energy it takes out of the batter head — but its own outward radiation leaves the far side of the shell and is kept as a separate diagnostic. Summing it at the batter microphone's point, phase, distance and polarity would require a propagation and diffraction model this reduction does not have, and adding it without one would be arithmetic rather than acoustics.

The mode coupling is truncated to one generation from a few pumps. Equation 13 alone is mean-field — every mode detuned by the same relative amount, **no mode able to transfer energy to any other**, which is the defining property of the Berger and Kirchhoff–Carrier family [4] and which would mean the nonlinearity contributed pitch and no spectral content whatever. Section 2.5 is that exclusion lifted, and what remains excluded is the size of the retained tensor rather than the mechanism. Full von Kármán coupling is $O(N^4)$ in coefficients and $O(N^3)$ to evaluate, so at 96 oscillators it arrives needing a truncation of its own, and the shipped one is severe: every retained term carries a pump index, the pump set is four modes below 700 Hz on the **batter** head only, and 256 of 408 structurally non-zero coefficients survive the budget. So there is one generation of transfer and no cascade — energy reaching a receiver cannot be re-radiated by it into a third mode — no parametric sidebands from the time-varying tension, and no coupling on the resonant head at all. The parity is not a truncation but a property: the force is cubic and odd, so $2f_i$ and $f_i \pm f_j$ are absent because they need a **quadratic** term in the potential, which a shell or a curved plate has and a flat tensioned head does not. That a hard hit is **brighter** and not merely sharper is measured in [10], a source cited here already for contact time, and again in [11], which tracks a tom-tom's modes across 67 strike intensities and resynthesises that evolution well enough that twenty listeners scored exactly chance in an AB test against the recordings — so what a strike does to the modes, and not only to the level, is what an intensity-dependent

Symbol	Quantity	Batter	Resonant	Unit
R	radius	0.1524	0.1524	m
σ	surface density	0.35	0.25	kg/m²
T	tension	1250	1040	N/m
D	bending stiffness	0.001	0.0007	N· m
c	wave speed $\sqrt{T/\sigma}$	59.76	64.50	m/s
f_{01}	fundamental	150.10	161.99	Hz
d_0	loss floor	0.8	1.0	1/s
d_1	constant-Q loss	0.4303	0.4644	m/s
d_2	excess HF loss	1.9e-5	1.9e-5	m²/s
d_{rad}	radiation loss	1.5	1.5	1/s
Δ_{01}	$(0, 1)$ correction	24.6	26.4	1/s
s	asymmetry split	0.004	0.003	—
β	tension coefficient	9.6e6	6.4e6	N/m³
	oscillators, Standard	96	24	—

Table 4: The two membranes at the shipped default. The resonant head is thinner and slacker, and carries only the modes the enclosed air can reach — a selection that is exact rather than approximate, followed by a truncation to its own budget that is not.

Quantity	Value	Unit
Cavity		
depth L	0.20	m
volume V	1.459e-2	m³
rigid $\rho c^2/V$	9.707e6	Pa/m³
stiffness scale s	0.083	—
pressure loss λ	5	1/s
A_{01}	3.150e-2	m²
air states	6	—
resonant mode limit	24	—
Nonlinearity		
tension ratio r	0.2	—
anti-alias bound	0.2346	—
Coupling		
$\tilde{\beta}$	7.0e5	N/m
pumps $\lvert P\rvert$	4	—
pump band	700	Hz
coefficients, of 408	256	—
worst force frequency	2709	Hz

Table 5: The air, the tension cap and the coupling channels at the shipped default.

drum sound is made of. This model produces part of that brightening as a source term and attributes the rest to excitation and to the attack layer's velocity scaling. Nonlinear modal synthesis with the coupling terms retained wholesale runs in real time [12]; this one, at the shipped truncation, does not yet — see Section 2.10.

There is no shell, no bearing edge, no vent and no hardware. These are real and audible on real drums; they are excluded because none of them can be calibrated against anything currently available, and a free parameter with no measurement behind it is indistinguishable from a fudge factor. There is no room and no snare. The cavity now has transverse modes but no **axial** ones, so a pressure that varies along the shell — and with it the sign difference between what the two heads see — is still absent, and so is any shell mode the air could couple to. The vent's absence is now quantified rather than asserted: it is a Helmholtz high-pass worth a few per cent of the head's volume velocity at the fundamental and less above, which is why it is excluded and also why excluding it does not explain the fitted stiffness scale.

And the reference set is synthetic. The committed fixture that pins this model's behaviour is generated from the model itself, deterministically. It is a regression reference, not an acoustic validation reference, and no measurement in this chapter should be read as agreement with a real drum. Where a number here does rest on a measured instrument — the contact time, the damping ratio, the cavity split — it is cited to the literature it came from, and Table 3 is where that is visible at a glance.

Two rows of that table lean on measurements that do not appear to exist, and both are recorded here as OPEN rather than guessed at. The first is the contact duration. Every source behind it is a **stick**: Dahl's 5.5–8 ms endpoints [10] and Wagner's crescendo [8] are both struck with sticks, and no measured felt-mallet contact time on any drum was found. The shipped mallet is therefore a stick's contact law given a different mass and hardness, and the longer, softer contact a felt head should give is uncalibrated — which matters because Equation 21 turns duration directly into the excitation spectrum. The second is the loss split. Equation 19 keeps radiation apart from the structural channels precisely so that calibration can attribute a change to a cause, and Table 4 assigns

Quantity	Value	Unit
Strike and contact		
radius, angle	0.30, 0.2	—, rad
footprint a_c	0.01	m
mallet mass	0.015	kg
contact stiffness K	1e6	$\mathrm{N/m^{1.5}}$
exponent α	1.5	—
hysteresis h	0.3	s/m
Microphone		
radius, angle	0.65, 0.6	—, rad
distance d	0.03	m
near-field scale s_{nf}	1	—
high-pass, low-pass	35, 12000	Hz
Attack layer		
level	0.05	—
placement	4000	Hz
band Q	0.7	—

Table 6: Excitation, observation and the stochastic layer.

$d_{\mathrm{rad}} = 1.5$ 1/s, but no numeric radiation-versus-internal damping split has been published for **any** drum. Only the total the two sum to is answerable to the constant-Q measurement [5] the law is shaped against; the division between them is a modelling choice wearing the notation of a measured one. Neither gap is filled by a plausible number here, because a number with nothing behind it would be indistinguishable in Table 3 from one of the measured rows.

3. THE TARGET

The target is one recorded tom hit. Its provenance is a sample library, with no documented microphone, room or processing chain, and no licence permitting redistribution. It is therefore a **timbre-match target**, not an acoustic validation reference: no automated test depends on it, and it is not part of the repository. Keeping that distinction sharp matters, because some of what the reduction reveals is a property of the recording rather than of any drum.

4. REDUCTION: FROM A SIGNAL TO NAMED QUANTITIES

Reference and candidate pass through the same extraction, so any asymmetry between them is a property of the signals rather than of the measurement.

Partials come from prominence-based peak picking on a 64k transform of the sustain window. Prominence rather than a bare local-maximum test, because ripples on the fundamental's skirt satisfy the latter and are not modes. Each surviving peak carries a level, a ring time from a least-squares fit to its log envelope, and that fit's R^2 — which later lets a partial's frequency count while its decay does not, since a beating pair or a mode buried under its neighbour has a slope but not a meaningful one.

Spectral envelope is a 24-band third-octave vector computed over each of four windows spanning the hit — attack, early, body, tail — so it describes not only the timbre but the way the timbre evolves. **Amplitude envelope**, **pitch glide** and **attack balance**, the click-to-body ratio, complete the reduction.

Deliberately absent is any sample-aligned waveform comparison. Against a recording of a different physical drum, waveform error measures the phase relationship between two signals that were never meant to share one: it is large for candidates that sound identical and small for candidates that do not. Its proper use, regression between two renders of the same model, is kept elsewhere.

5. THE DISTANCE

5.1. Units and weights

Each term is reported in the unit its own perception is measured in: partial frequency in cents, because the ear hears pitch as a ratio and 3 Hz means something different at 118 Hz than at 1180; partial decay as $\left|\ln\left(T_{60}^{\mathrm{ref}}/T_{60}^{\mathrm{cand}}\right)\right|$, because ringing twice as long and half as long are the same size of mistake; levels and envelopes in dB; glide in cents.

Weights are the reciprocal of each term's "clearly wrong" threshold — 25 cents of pitch, 3 dB of partial balance, a factor of 1.4 in ring time, 4 dB of spectral shape, 3 dB of envelope, 40 cents of glide, 6 dB of attack balance — so a just-audible error anywhere contributes about equally to the sum. Both signals are peak-normalised, making every term gain-invariant; that is forced rather than chosen, since the reference is normalised and its loudness carries no information.

5.2. Matching

Partials are matched greedily by closeness in cents rather than in index order, so one badly placed candidate cannot cascade a misidentification through the series, and each candidate is claimed at most once, so it cannot explain two reference modes. The tolerance widens with mode index, since real two-headed drums scatter about $\pm 20\%$ around the ideal Bessel series in both directions [2].

5.3. Coverage, in both directions

Six terms are computed over matched pairs, and an error averaged over matched pairs is zero when there are no pairs. Each is therefore blended against a fixed penalty in proportion to the share of the reference left unmatched, so that a partial which is missing costs what a partial which is present but wrong costs.

That share must be weighted, and the weighting is the substantive choice. Weighting by **energy** concentrates it on whichever partial is loudest — on this reference, one partial carries 99.4% of the total — so a candidate holding that partial alone reports almost nothing missing. Weighting by **count** overcorrects: the reference contains a genuine, isolated component 41 dB down that no two-headed drum will produce, and failing to reproduce it must not cost what failing to reproduce the fundamental costs. Each partial is therefore worth **how far it stands above the detection floor, in dB** — zero at the floor, growing with prominence, monotone in loudness like energy but compressed enough that several missing quiet partials cannot be rounded away.

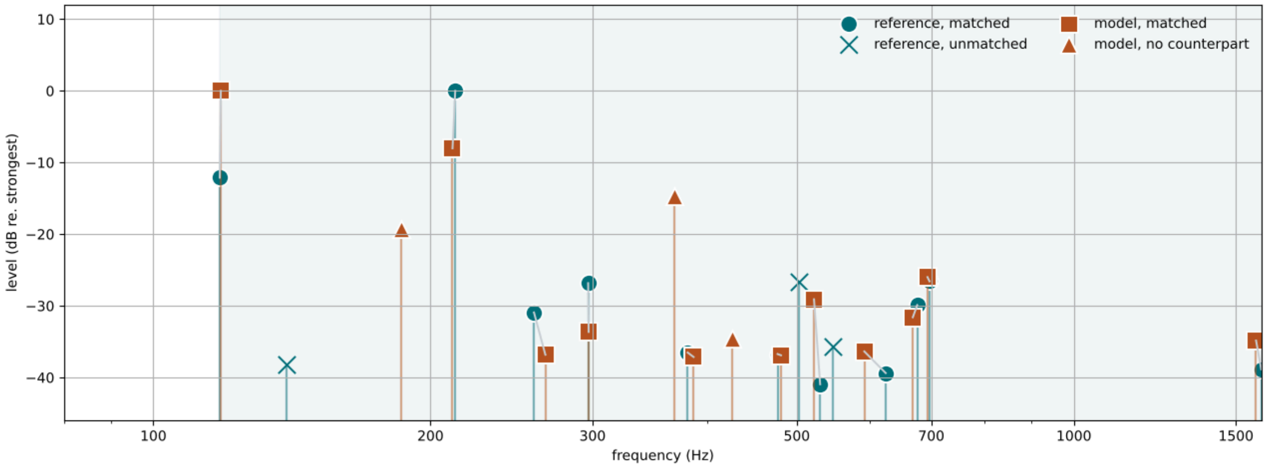

Figure 6: What the two coverage terms see. Grey links join matched pairs; shading marks the span between the lowest and highest reference partial, within which the spurious share is counted. Eleven of the reference's fourteen partials find a counterpart. Of the three that do not, two sit near the detection floor and are correspondingly cheap to miss, while the third stands 26 dB below the strongest and carries most of the unmatched share on its own — which is the audibility weighting doing exactly what it is for.

A second share scores the mirror case. Because matching iterates the reference, a candidate partial with no reference counterpart is invisible to every partial term and reaches the sum only through the spectral envelope, so a candidate can cover the reference completely while its second-loudest component is a mode the reference does not have. Both shares therefore appear, under the same weighting and at the same weight. Figure 6 is the picture they score.

The spurious share is counted **only** between the lowest and highest reference partial. Outside that span the reference's own detection is unproven — a recording's noise floor hides modes a model legitimately has — so a partial out there is charged by the spectral envelope, on evidence, and not by coverage. Without that bound the term would fit the recording's limitations rather than the drum.

5.4. The gate

The sum orders candidates; it does not decide them. Adoption is gated on three terms individually — partial frequency within 25 cents, partial decay within 0.25 in log ratio, spectral envelope within 4 dB — because a lower sum can be bought in terms nobody listens for.

That is not hypothetical. In one run the candidate with the **best** spectral envelope of any measured had the **fewest** partials, three, having found that raising the stochastic attack layer's level and dropping its corner frequency into the band of interest satisfies a band-energy measure. Broadband noise can satisfy a spectral envelope; it cannot produce resolvable partials. On the total alone that run looked like the best match to the recording.

6. PREPARING THE REFERENCE

A stereo recording of one hit is two views of the same event, and they are generally not simultaneous. Summing them is then a comb filter: for a delay τ,

$$H(f) = 2|\cos(\pi f \tau)| \tag{27}$$

with nulls at $(2k+1)/2\tau$ and maxima at k/τ — a fixed ± 6 dB spectral shape imposed by microphone geometry alone [13].

The reduction therefore measures the inter-channel lag by cross-correlation and aligns before averaging. It fires only when the pair is genuinely one signal twice — correlation at least 0.5, and clearly better than summing — and never on short or uncorrelated input. The measured lag and correlation are reported on the decoded reference, so the condition cannot pass unnoticed, and regression tests pin both directions: a delayed pair is aligned, an uncorrelated pair is left alone.

On this recording the second channel lags the first by 69 samples at 44.1 kHz, 1.56 ms, with correlation 0.942 at that lag against 0.361 at zero. Equation 27 then predicts nulls at 320, 959 and 1598 Hz and maxima at 639 and 1278 Hz, which is what the naive downmix shows against a single channel, with no fitted parameter (Figure 7).

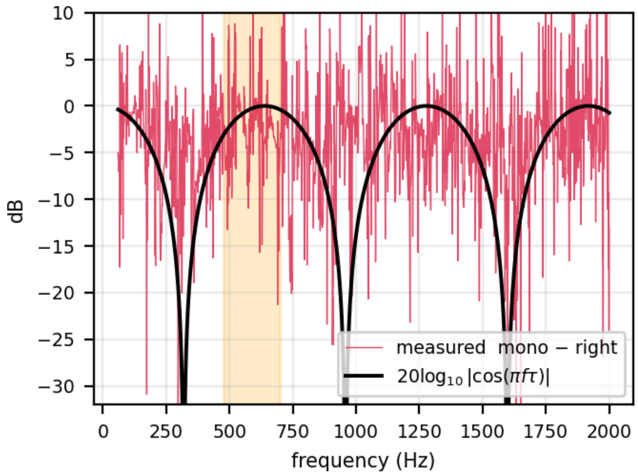

Figure 7: The naive mono downmix relative to the right channel alone, against Equation 27. Shaded: 476–700 Hz, where the comb's first maximum falls.

The effect on the reduction is large, because the partial detector keeps a bounded number of peaks and the comb decides which survive:

Reference reduced as	Partials	In 476–700 Hz
mono, averaged naively	16	9
right channel alone	7	2
mono, aligned then averaged	**5**	**2**

Table 7: One recording, three reductions. Aligning before summing brings the mono view into agreement with the single channels; the naive sum agrees with neither. Counts are as measured before the level estimator of Section 11 was corrected; the corrected right-channel reduction finds 14 partials, 7 of them in 476–700 Hz. The agreement the table reports is between reductions measured the same way, which the correction moves together.

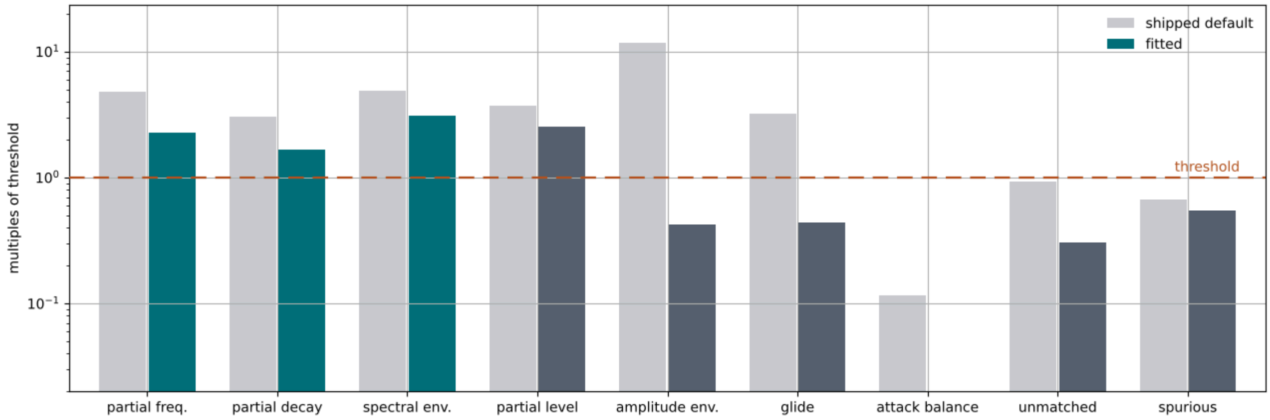

Figure 8: Every term as a multiple of its own threshold; the three gated terms are highlighted. Five terms are brought below threshold, four remain between 1.7× and 3.1×, and the two coverage shares confirm the candidate is a drum rather than either degenerate extreme.

Aligning also brings the reference's ring times into agreement with membrane physics. Constant Q requires $T_{60} \propto 1/f$ [5]; the naive downmix gives a median of 1.13 s in the 476–700 Hz band where that law predicts 0.31 s, a factor of 3.6 that invites explanation in terms of a reverberant room. The aligned reduction gives 0.16, 0.19, 0.60 and 0.61 s, which needs no such explanation.

7. THE SEARCH

The distance is minimised with the Mayfly Optimization Algorithm [14], a build-time-only dependency imported by the fitting command and nothing else — the shipped binary is unchanged. The search space is exactly the bank the product exposes, plus strike velocity, with the quality tier pinned: mode count buys fidelity with CPU and is a product decision rather than a property of this drum.

Runs are deterministic given the seed. Progress is checkpointed with the completed restarts and the running best point, written atomically, and the checkpoint carries a fingerprint of the baseline cost. A resume across any change to the model or the measurement is refused rather than silently mixed — which has repeatedly caught a stale checkpoint that would otherwise have produced a run whose early generations were scored by one metric and its later ones by another.

One property of the model is worth stating here, because it determines the tier a comparison must run at. Mode selection retains a fixed **count** — 48, 96 or 160 by quality tier — while membrane mode frequencies scale as $\sqrt{T}$. Rendered bandwidth is therefore a function of tier **and tuning** together, and it falls as the drum is tuned down. For a bank at 331 N/m, roughly a quarter of the shipped batter tension:

Tier	Modes	Top mode	In 476–700 Hz
Draft	48	467 Hz	0
Standard — *shipped default*	96	664 Hz	45
High	160	852 Hz	58

Table 8: Modal ceiling of a low-tuned bank, by quality tier.

Two consequences follow: an instrument truncated by count loses its top octave when it is tuned down, and any comparison between configurations must run at the tier that ships. Bounding mode selection by frequency instead would make the ceiling independent of tuning.

8. WHAT THE FIT ESTABLISHES

Read with Section 12. The glide term of the distance minimised below was later found to be faulty, and both the glide rows in this chapter and the search that produced the banks are conditioned on it. The numbers here are reported as they were obtained; Section 12 gives the fault, its size, and what of this chapter it puts back in question.

The run reported here fits against the right channel at the shipped Standard tier with the prescribed contact model, under the corrected reduction of Section 11: eight restarts of 150 iterations over a population of 16, seed 1, four of the eight restarts seeded from the pre-solve of Section 10. All eight completed, 59,056 evaluations, and everything below is read from the report it wrote.

The distance falls from 33.094 at the shipped default to 11.252. More usefully, Figure 8 shows where it fell.

Five of the nine terms are brought below threshold: amplitude envelope to 0.42 × (1.27 dB against 3), glide 0.44 × (17.6 cents against 40), attack balance 0.004 × (0.03 dB against 6), and both coverage shares — unmatched 0.30 ×, spurious 0.55 ×. The coverage pair is the useful confirmation: the candidate accounts for most of the reference's audible partial content while inventing little of its own, so the remaining terms are computed over a genuine correspondence rather than over one lucky pair. The reference reduces to fourteen partials and the candidate to sixteen; eleven pair up, and the three reference partials left over are what the unmatched share prices (Figure 6).

Four terms remain above threshold — partial frequency 2.28 × (56.9 cents against 25), partial decay 1.66 × (0.58 in log ratio against 0.35, and against the tighter 0.25 its own gate applies), partial level 2.53 × (7.58 dB against 3), spectral envelope 3.07 × (12.28 dB against 4) — and they are not four independent problems. Figure 9 and Figure 10 show one.

The remaining error is a shape of the loss law. Ring times do not sit uniformly above or below the reference's: at the fundamental the candidate rings 1.13 s against the reference's 1.49 s — too short — while between 290 and 390 Hz it rings 1.4 to 2.2 times too long, and across 370–700 Hz it delivers a nearly flat 0.6–1.0 s where the reference ranges from 0.26 to 1.20 s. The same fact appears in the spectral envelope as an excess in the later windows above about 1 kHz, energy that should have decayed and has not, with the attack window matching within a few dB. Decay, not spectral shaping, is what the gated terms are measuring.

The reference's own decay times are strikingly non-monotone in frequency: its loudest partial at 213 Hz dies in 0.146 s while its

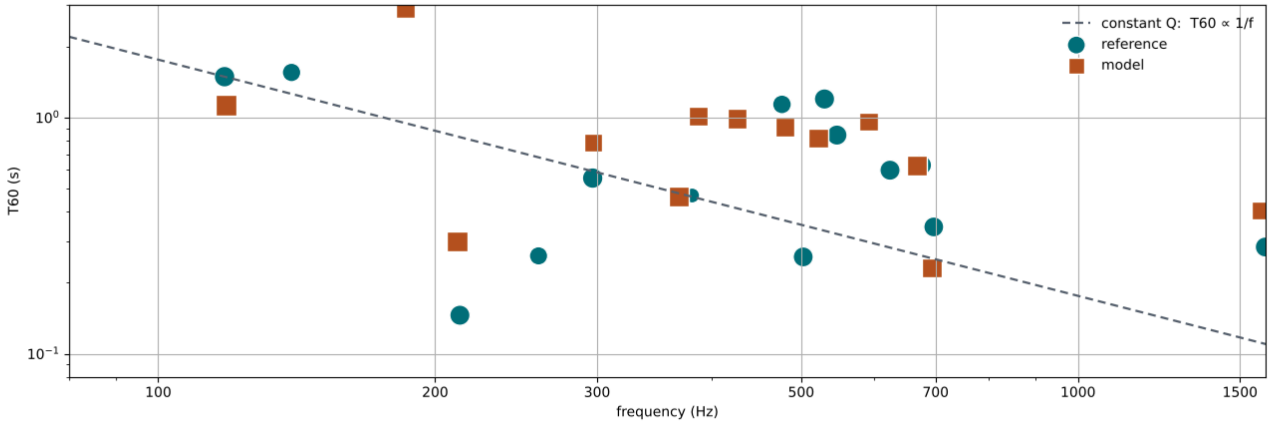

Figure 9: Ring time against frequency, with the constant-Q law anchored on the reference fundamental. The mismatch is a shape rather than an offset: the model's ring times bunch between 0.6 and 1.0 s across 370–700 Hz, where the reference scatters over a factor of four, and it rings about three-quarters as long as the reference at the fundamental.

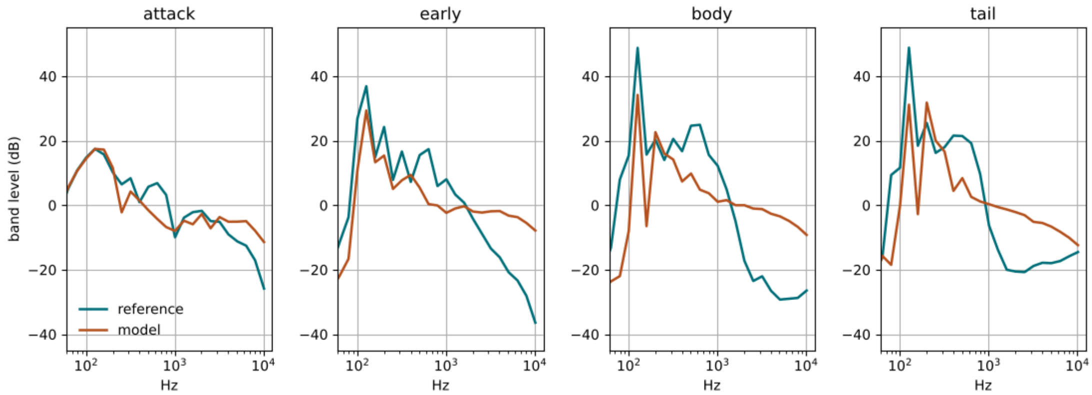

Figure 10: Spectral envelope, window by window. The attack matches within a few dB over most of the band; the discrepancy grows through the early, body and tail windows and is concentrated above about 1 kHz, where the model runs 10–25 dB hot.

118 Hz fundamental sustains 1.49 s. No single constant-Q loss law produces that ordering, which is why a fit that gets the mid band right must get the fundamental wrong, and the reverse. Locating the residual as a shape of the loss law rather than as a level is exactly the discrimination the per-term units were built for.

The same apparatus settles a design question. The prescribed contact model and the Hertzian alternative were fitted separately at identical budget, seed and reduction — the only difference being the flag — so the comparison is between two excitations over the same bank rather than between one excitation and a default that happens to suit it.

Term	Threshold	Prescribed	Hertzian
Partial frequency	25 cents	56.9	**47.2**
Partial decay	0.25	**0.581**	0.728
Spectral envelope	4 dB	12.28	12.30
Partial level	3 dB	**7.58**	9.18
Amplitude envelope	3 dB	1.27	**1.18**
Glide (see Section 12)	40 cents	17.6	**0.12**
Attack balance	6 dB	**0.03**	1.31
Unmatched share	0.5	0.151	**0.047**
Spurious share	0.5	**0.275**	0.359
Total		**11.252**	11.535
Baseline		33.094	33.544

Table 9: Two contact models at identical budget, seed and reduction. Bold marks the better of the pair on each term. The prescribed model wins the sum, and the two win on different terms — which is the reading the per-term units exist to make available.

The prescribed model takes the sum, 11.252 against 11.535, and the restart spreads say how firmly. Its eight restarts run 11.252 to 15.461 and the Hertzian's 11.535 to 18.904; the Hertzian best would rank second among the prescribed restarts while its remaining seven fall outside the prescribed range's better half. That is a **consistent** ordering rather than a decisive one, and it is worth stating in those terms: the prescribed excitation earns the default it already holds, and nothing here claims it is the better physics.

What the per-term reading adds is that the two do not merely differ in size. The Hertzian fit is better placed in frequency, has essentially no glide error and leaves less of the reference uncovered; the prescribed fit has the better decay, partial balance, attack balance and tidiness. A single number would report one winner and conceal that the two are good at different things — which is the information a next iteration of the excitation model would want.

A normalised position beside an engineering value turns a fitted parameter into a question. Head damping here is fitted to 0.709 at normalised position 0.376, comfortably inside its range. An earlier run under the superseded reduction fitted the same parameter to 0.276 at position 0.036 — against the edge — and it was the position rather than the value that made that legible. Section 9 puts that question to the model directly.

9. TESTING A BOUND WITHOUT WIDENING IT

A parameter resting on a bound invites a range change, and a range change is not free: presets store **normalised** positions, so widening a shipped spec silently retunes every saved patch. The question can be asked without touching the product. A build-time `-loss-scale` multiplier applies to every head loss rate on top of the head-damping parameter, so a run at 0.25 searches a range whose floor is four times lower while the shipped spec is exactly as it ships. It is not a product knob and is not intended to become one: a bank fitted at a multiplier other than 1 describes a drum the product cannot be set to. Both the report and the checkpoint fingerprint record the multiplier, so such a run can neither be mistaken for a normal one nor resumed into one.

Three fits at equal budget — 4 restarts of 60 iterations over a population of 12, seed 1 — identical but for the multiplier:

Scale	Baseline	Best	P. decay	Spec. env.	Damping
1.0 *(control)*	32.585	**14.917**	0.966	13.27 dB	0.475 (0.231)
0.5	26.388	**14.924**	1.263	13.01 dB	0.866 (0.448)
0.25	26.325	**14.076**	1.295	12.82 dB	2.122 (0.771)

Table 10: Equal-budget sweep of the loss multiplier; head damping is given as the fitted value with its normalised position in brackets. Effective damping — the product of the fitted value and the multiplier — is 0.475, 0.433 and 0.531: the same drum, reached from three different ranges.

Four times the headroom moves the total by 0.8, less than the spread between restarts at this budget, and the fitted normalised position travels 0.23 → 0.45 → 0.77 to compensate for the multiplier — which is what a parameter does when the model can already reach the value the reference implies. At this short budget the parameter does not rest on its bound even in the control, so the sweep alone speaks to the range and not to that particular pin. A full-budget run at the lowest multiplier settles it:

Scale	Restarts	Best	Damping	Effective
1	stopped at 47%	**11.630**	0.276 (0.036)	0.2764
0.25	all 8 complete	**13.023**	1.132 (0.545)	0.2830

Table 11: The same question at full budget, damping again with its normalised position in brackets. Given four times the range, the search chooses the same physical damping to within 2%.

Both tables are measured under the estimator Section 11 corrects, so the totals are superseded; what the experiment establishes survives it, because the comparison is between runs measured identically and the conclusion is drawn from the **agreement** between them rather than from any one value.

The bound is not binding, and the shipped range needs no change. The parameter comes off it as soon as the bound stops mattering, which is what a non-binding constraint looks like from the inside; the pinned value was a property of that basin rather than a limit of the bank.

What the experiment establishes positively is where the residual lives. Removing loss makes the partial-decay term **worse** — 0.966 to 1.295 across the sweep — while the spectral envelope sits flat near 13 dB and never approaches its 4 dB gate, so the envelope's excess is not made of over-long modes and damping will not remove it. Together with the non-monotone reference decay of Section 8, this places the residual in the **shape** of the loss law: head damping scales the whole law and the decay tilt slopes it, and a curve that falls and rises again is neither a scaling nor a slope. A per-mode decay correction, already scaled by both knobs, is where such an answer would go. That is a model-structure question, not a range question and not a search question, and separating the three is what the per-term units are for.

10. SEEDING A RESTART FROM THE REFERENCE'S PARTIALS

Mode frequencies are analytic: they are read off tension, radius and cavity without rendering a sample, at roughly a hundredth of the cost of one full evaluation. That makes it cheap to ask a question the fit itself cannot afford — how close can the model's modes get to this recording's partials at all?

Measurement	Audibility-weighted frequency error
20,000 random banks, best	11.5 cents
the same, hill-climbed	10.5 cents
converged pre-solve, corrected target	35.9–37.0 cents
gate	25 cents

Table 12: What the analytic pre-solve reaches, against what the gate asks for. The first two rows are measured against the reduction of the reference as it stood before the correction of Section 11; the third against the corrected reduction, whose fourteen partials include seven inside a 224 Hz span. The pre-solve is the same in both cases — it is the target that changed.

Read that table as a property of the **searchable space** rather than of any search: it reports how close the modes of the best bank the pre-solve can construct come to the partials it is asked to hit, before a single sample is rendered. Against the earlier reduction the answer was around a cent, well inside the gate. Against the corrected one it is around 36 cents, outside it. Obtaining that answer costs roughly one full evaluation, which makes it a cheap thing to know in advance and, as below, a decision procedure rather than only a diagnostic.

`-seeded-restarts N` puts it to use: a pre-solve finds N diverse frequency-optimal banks, and those N restarts search a box around them while the remainder search the whole cube. The seeded restarts come first and the unseeded ones keep their RNG seeds, so they are bit-for-bit what they were without the flag and the comparison is paired.

A seed is evidence about some parameters and not others, and the box has to respect that. A frequency-only objective has no opinion about damping, the microphone position or the attack layer, so narrowing a restart in those directions spends range for nothing. The dimensions the seed speaks to are therefore found by **probing** which ones move the mode frequencies, rather than by listing them, so the mask stays true when the parameter table changes. Here it selects 4 of 17 free parameters — overall size, the two head tunings and the head asymmetry, which are the wave speed, the two tensions and the mode split — and, against the reduction as it then stood, the pre-solve's two seeds land at 1.0 and 1.3 cents.

Restart	Control	Seeded, all dims	Seeded, masked
1 — *seeded*	19.442	31.707	**16.388**
2 — *seeded*	16.754	19.109	**13.132**
3	14.917	14.917	14.917
4	16.638	16.638	16.638
Best	**14.917**	**14.917**	**13.056**

Table 13: Paired comparison at equal budget. Restarts 3 and 4 are unseeded in all three runs and reproduce exactly, which is what makes the first two rows readable.

With the box restricted to those four dimensions, both seeded restarts improve and neither regresses: a 12% better result at the same cost. The partial-frequency term falls from 94.4 to 46.1 cents, partial decay from 0.966 to 0.766, and the unmatched coverage share from 0.461 to 0.011 — the seeded drum produces the reference's partial content rather than a fraction of it. The spurious share rises, 0.413 to 0.638, which is the honest price of the method: a bank chosen to have a mode near every reference partial also has modes elsewhere, and Section 5.3′s second share is what reports it.

A seed helps in proportion to how good it is, and the pre-solve says how good it is before the search starts. Table 13 is a 1-cent seed, and it buys 12%. The corrected reduction of Section 11 gives the same pre-solve a harder target, its seeds floor at 36 cents, and the two full-budget fits of Section 8 — eight restarts each, four of them seeded — were both **won by an unseeded restart**: 11.252 against a best seeded 11.741 with the prescribed contact, 11.535 against 13.574 with the Hertzian. A box is range spent to buy a head start. At one cent the head start is a real selection from the distribution below and the trade is worth making; at thirty-six the seed is barely a selection, and the box only removes room. The flag therefore stays off by default, and the number that decides whether to raise it is printed by the pre-solve for the price of one evaluation.

That is the durable part of this section. Table 13 and the cent figures in it are measured against the earlier reduction and do not transfer, but the mechanism, the two defects the first version exposed, the paired design that makes the table readable, and the rule above all survive the change of target — and the rule is only visible **because** the method was tried against two targets of very different difficulty.

11. THE CONDITIONING OF A LEVEL ESTIMATE

The reduction attaches a level to each partial, and coverage, matching and the detection floor all depend on it. That estimate was obtained exactly, and probing it showed the exact formula to be an unusable estimator. The finding is worth reporting for its own sake: **conditioning is a property of a measurement independent of its correctness**, and an apparatus built to be read term by term is exactly the kind that can be probed to expose it.

The estimate was the partial's magnitude in the sustain transform divided by the attenuation that window applies to a partial decaying at the fitted rate. For an isolated exponential that is exact. But the sustain window spans 0.05–0.85 s, and the divisor it implies grows without bound as the fitted ring time shortens:

T_{60} (s)	2.0	1.5	1.0	0.6	0.3	0.15	0.073
Correction (dB)	12.4	16.1	22.9	34.2	54.9	82.3	122.0

Table 14: What the window correction asks of the measurement. Across the ring times this recording actually contains, the divisor spans more than 100 dB.

A quantity recovered by dividing through Table 14 is largely a restatement of the fitted decay rate: at 0.12 s a 10% error in T_{60} moves the reported level by 4.5 dB. On this reference the consequence was visible. At a 5 Hz peak separation a 73 ms component — T_{60} 0.073 s, R^2 0.865 — was reported as the loudest partial in the recording, which put the fundamental — T_{60} 1.49 s, R^2 0.984 — at −33.6 dB. Levels are relative to the strongest, so genuine partials then fell below the detection floor and disappeared, and the detected set was not monotone in that floor.

Capping the correction is the obvious remedy and the measurement rejects it: the reference's genuinely loudest partial, at 213 Hz, itself carries an 83.7 dB correction, so a cap tight enough to exclude the runaways excludes it too.

The estimator is instead replaced by a directly measured one. The per-partial decay fit is a least-squares line through the log of the partial's heterodyned envelope, and its **intercept** — the fitted line's value at $t = 0$ — is the same quantity the correction was reaching for. It extrapolates back only as far as the start of the fit window rather than through the whole taper, and it rests on a fit over hundreds of samples rather than on a ratio of two numbers. One constraint carries over intact and is respected: the level must not be the envelope's **peak** inside the fit window, because a strike transient smeared through a 150 ms filter once placed the 213 Hz partial 32 dB too loud. The fit begins after the transient, and the intercept is an extrapolation of the fitted decay rather than a reading of the envelope.

A second detection window follows from the same arithmetic. A partial ringing a tenth of the 800 ms sustain window stands roughly 90 dB lower in its transform, and both detection guards — a relative floor with 20 dB of headroom, and a count limit — rank on that uncorrected magnitude. They bind together, which is why loosening either alone establishes nothing: sweeping the headroom from 20 to 80 dB left the count at 7 throughout, and so did sweeping the count limit over a factor of sixteen; only both at once moved it. Detection therefore reads a second, earlier window spanning 0.05–0.30 s, and **each window admits candidates relative to its own strongest peak**, so a short-ringing partial competes against short-lived content rather than against the fundamental's entire ring. The sustain window is admitted first, so where both see a partial the better-resolved frequency is kept; the early window starts after the transient, since a broadband click offers a peak at every frequency. The shipped guard constants are unchanged.

The order of the two fixes is the transferable part. The tight guards were, in effect, what had been keeping the unstable estimator from firing: they excluded precisely the short-ringing population where its correction ran away. Opening the aperture first would have made the measurement worse. Bound the estimator, then widen what it is asked to see.

A regression test pins both on a synthetic two-partial signal — amplitude 1.0 ringing 1.2 s against amplitude 0.5 ringing 0.12 s — so the behaviour is fixed without reference to any recording. It failed twice while being written: first because the levels came out inverted, then because the short partial was not detected at all.

Right channel, reduced	Partials	In 476–700 Hz
before the correction	7	2
after	**14**	**7**

Table 15: The corrected reduction of the reference. Half of what it now reports lies in the band the recording's character lives in.

Table 15 re-scopes a standing claim, and the fits of Section 8 then narrow it. Fits under the earlier reduction reproduced little in 476–700 Hz while the reference was busiest there, and the reference still is — but part of that gap was the target not asking for those partials either, because they sat below the floor the level estimate set. On the corrected metric both full-budget fits place **five of their sixteen partials** in that band against the reference's seven, so the model does populate it. Whether the remaining difference in density is an excitation deficit or a mode-density one is open, and the corrected reduction is what makes it answerable: the target now asks for the partials the question is about.

One quantity to keep in view while answering it: the analytic pre-solve of Section 10 floors at 36 cents against this target, so the frequency gate's reachability within the shipped bank is itself an open question. The gate was calibrated against a reduction with half as many partials, and a per-partial cent tolerance does not ask the same thing of a model when seven partials must be placed inside a 224 Hz span. Re-deriving it against the corrected reduction is the natural next step, and it is a question the apparatus can now be pointed at directly.

12. A GLIDE READ OFF A DEAD PARTIAL

Section 11 is a measurement defect found by probing, corrected, and then folded into the fits it changed. This chapter is the same genre of finding with the opposite disposition: the defect was found **after** the fits of Section 8 were run, and it is reported here as a correction **to** them rather than absorbed into them. Nothing above this chapter has been restated. The paper is a record of what was done, and a run described in one chapter and silently re-scored in another is not one.

The glide term reduces the whole pitch bend to a single number: the frequency of one tracked partial at an early probe against its frequency at a late one, in cents. Two faults, independent of each other, were in that estimator throughout the work reported above.

The late probe fired after the tracked partial was dead. The probe sat at a fixed 0.400 s. The model's $(0, 1)$ has a ring time near 0.21 s, so by then it is roughly 105 dB below where the early probe found it, and what the second reading returns is not that partial at all but whatever leaks through the probe's passband from a neighbour — reported, in cents, as the offset to that neighbour. On ordinary renders this produced readings such as −717 cents. At a weight of one part in forty cents that is a contribution of 18 to totals of order 32: a term nominally worth 0.44 in Table 9 could, on a different candidate, be worth more than half the distance.

The glide was read off the loudest partial rather than the fundamental. The loudest partial was chosen deliberately, and for a reason that still holds — the lowest peak in the band may be a shell resonance or a room mode forty decibels down, and the bend belongs to the mode carrying the note. But on this reference the loudest partial is at 212.74 Hz with a 0.162 s ring time, while the fundamental at 118.05 Hz is 7.7 dB quieter and rings for 1.53 s. The measurement was being taken on whichever mode peaked highest rather than on the one that still existed to be measured.

Fault two is the one with reach, because it moved **the target**. The reference's glide is not what the fits of Section 8 were asked to match:

Reference reduction	as measured then	corrected
Mono	89.3 cents	**54.1 cents**
Right channel, as used in Section 8	120.4 cents	58.9 cents

Table 16: The reference's downward glide before and after the correction. The earlier figures were read from a mode that had decayed into the noise floor by 0.15 s; the corrected ones track the fundamental, whose trajectory over the same span is monotone.

The estimator now walks the late probe back from its nominal position to the last point at which the tracked partial is still within 20 dB of its early level, refuses any early-to-late span shorter than 50 ms outright, and tracks the lowest partial standing within 20 dB of the loudest — which keeps the guard against the forty-decibel room mode while preferring the fundamental, on a drum the longest-lived thing in the recording. It also returns a **measured** flag beside the value, so that a take whose fundamental cannot support two probes is reported as unreadable rather than as zero cents. That last part is the piece worth generalising, and it is the eleventh point of Section 14.

12.1. What this corrects, and what it does not

The correction reaches further into Section 8 than a rescored row. The glide term was inside the objective the search minimised, so both fits in Table 9 are **conditioned** on it: it is not that nine terms were measured correctly and one was not, but that the optimiser was steered, at every evaluation, by a term that could be worth 18 on a candidate whose fundamental died early. The fitted banks of Section 8 are outputs of the faulty objective and are described as such.

Two specific consequences are worth naming rather than leaving to the reader.

The contact-model ordering is not established by Table 9. The margin is 0.283 and the prescribed fit's glide row contributes 0.44 against the Hertzian's 0.00, both measured against a target of 120.4 cents that should have read 58.9. The ordering may well survive — it is a consistent one across restarts — but it is no longer supported by the numbers printed. Section 13 re-derives the prescribed side under the corrected objective; the Hertzian side has not been re-run, so the ordering itself is still outstanding.

The shipped Berger β is untouched. It was fitted to the model's own glide on an isolated batter fixture — 96.9 cents between velocity 0.2 and velocity 1.0, 104.9 at the shipped coefficients — measured directly from mode frequencies and never through this estimator or against this recording. The **pending refit** recorded against it in Table 3 is the coupling question of Section 2.5 and is unrelated to anything here.

What is worth reading against it instead is the corrected target itself. Three independent measurements put a loud hit on a real tom between 130 and 165 cents — Fletcher and Rossing's §18.4 relays Bork and Meyer at 160 cents on a 32 cm tom and Rose at about 8% on a 33 cm one [5], [15], and Gärder measures almost 8% on a 14-inch drum [16] — against this reference's corrected 54.1. Those do not conflict; the natural reading is that the reference is simply not a loud hit. But the reference is what the objective is written against, so a nonlinearity fitted through it is fitted to match **this recording** and should be read as that rather than as a physical coefficient.

12.2. A parameter the recording cannot see

One further result falls out of the correction, and it is the kind that is easier to state than to obtain. Swept over the cavity stiffness scale s from the shipped value to a rigid shell, the objective total used to range 32.83 to 53.15 — a spread of 20.3, which reads as strong discrimination. Under the corrected estimator the same sweep ranges 32.98 to 34.10, a spread of **1.1**.

Almost all of the apparent discrimination was the glide artefact: at the strongly coupled end the tracked partial died early and the probe read a neighbour, and the size of that misreading varied with the coupling. What remains says that **this recording has essentially nothing to say about s**, which is a substantive finding given that s is the one cavity parameter the model fits rather than derives (Table 3) and that its value is under independent question from the literature. A parameter that cannot be identified from the target is a parameter whose value has to be settled by a measurement made for the purpose — Fischer's doublet protocol on a tom — and not by a better fit.

13. THE REFIT UNDER THE CORRECTED OBJECTIVE

Section 12 leaves a debt: the banks of Section 8 are outputs of an objective that has since been corrected, and re-deriving them under the corrected one was recorded there as outstanding. This chapter pays part of it. The prescribed contact model was refitted from scratch, at a budget larger than Section 8 ran at, against the corrected estimator. The Hertzian alternative was **not** refitted, so Table 9 is not superseded here — it remains a superseded result, and the contact-model ordering remains undetermined.

13.1. Putting two objectives on one axis

An archived report's stored total was produced by an objective that no longer exists, but the report records its own best point exactly, so the bank can be re-measured: pin every free parameter to its

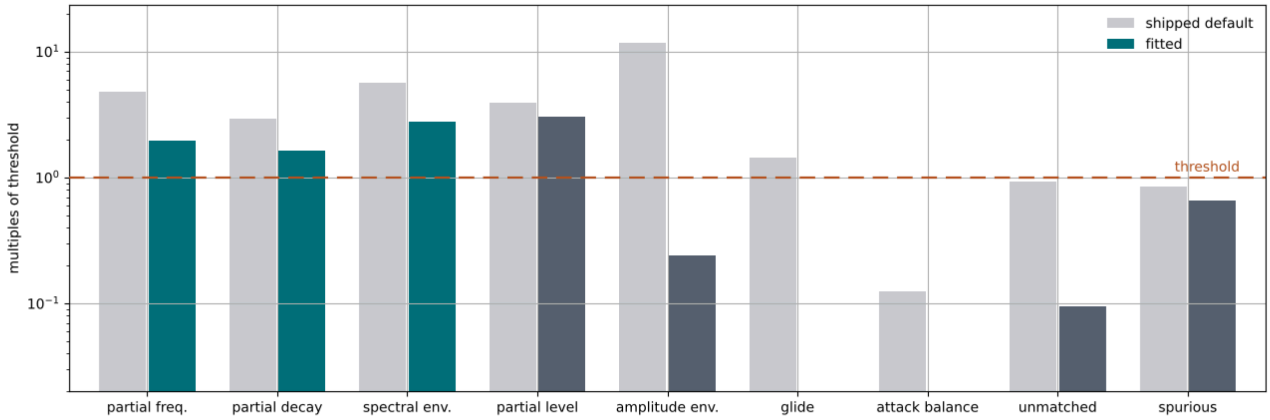

Figure 11: The refit, every term as a multiple of its own threshold, drawn as Figure 8 is. Glide and attack balance have no visible fitted bar because they fall off the bottom of the axis — 0.03 cents against a 40-cent threshold and 0.02 dB against 6. The three gated terms are highlighted and all three are still missed.

recorded normalised position, pin the strike velocity, and ask for a report without a search. Quality tier, contact model and channel must be reproduced from the report's own header, or the answer is about a different drum.

Archived run	Then	Today	Glide (cents)
Control, post-fix	10.577	**10.577**	1.9 ⟶ 1.9
Hertzian, Table 9	11.535	15.146	0.1 ⟶ 59.6
Hertzian, earlier	11.630	15.650	— ⟶ 49.7
Prescribed, Section 8	11.252	**18.991**	17.6 ⟶ 107.1

Table 17: Four archived banks re-measured under the corrected objective, each at its own tier, contact model, channel and velocity, against their totals as reported at the time. The first row is the recipe validating itself: it is the one archived run made after the correction, and re-scoring reproduces its reported total to the digit, term for term.

Two readings of that table are wrong and worth closing off. It is **not** a measure of what the glide fault alone cost, because a re-score renders the bank through today's synthesis as well as measuring it with today's metric, and the model gained its mode-to-mode coupling channels (Section 2.5) on by default in the same interval — the shipped baseline moved 33.094 ⟶ 32.442 on the same bank and channel for that reason alone. And it is **not** a ranking of the searches that produced those banks, because each was optimised against a different objective from the one marking it. What it does say is that the run which won under the old objective is the worst of the four under the corrected one, and the glide column says why: a near-zero error against a fabricated target is a bank tuned onto the artefact.

13.2. The run

Twelve restarts of 150 iterations over a population of 16, four of the twelve seeded from the pre-solve of Section 10, seed 1, right channel, Standard tier, prescribed contact. All twelve completed; 88,584 evaluations. **The distance falls from 32.442 at the shipped default to 10.382.**

Five terms again fall below threshold and the same four remain above, but the comparison that matters is not with Section 8, whose total is not on this axis. It is with the one archived run measured the same way — the control row of Table 17, an interrupted eight-restart run at a different seed.

Term	This run	Control
Partial frequency	48.936 cents	48.997 cents
Partial level	9.081 dB	9.172 dB
Partial decay	0.573	0.602
Spectral envelope	11.068 dB	11.034 dB
Amplitude envelope	0.724 dB	0.746 dB
Unmatched share	0.047	0.047
Spurious share	0.327	0.330
Glide	**0.029 cents**	**1.892 cents**
Attack balance	**0.019 dB**	**0.188 dB**
Total	**10.382**	10.577

Table 18: Two independent searches under the same objective — twelve restarts against eight, different seeds, one complete and one interrupted. Every term the objective can read about the drum agrees; the whole of the 0.195 difference is the two bold rows, which are the cheapest terms in the sum to polish once the rest has settled.

That agreement, and not the 1.8% better total, is the result. Two searches arriving at the same point from different starts is a statement about the model's ceiling on this recording rather than about either search: the remaining 10.4 is not search effort left on the table. The winning restart's convergence agrees from the inside — its last three iterations read 10.3829, 10.3823, 10.3821, flat to four digits, where the run of Section 8 was still visibly descending when its budget ran out. A better total on this reference now needs a different model, not a longer run.

13.3. What the ceiling is made of

All three gated terms are missed, for the third round running: partial frequency 48.9 cents against 25, partial decay 0.573 against 0.25, spectral envelope 11.07 dB against 4. The spectral envelope is now on its **sixth** independent intervention — contact model, head-damping range (Section 9), seeding (Section 10), the level correction (Section 11), the glide correction (Section 12), and a converged full-budget refit — and it has sat between 11 and 13.6 dB throughout. A term that does not move for six unrelated changes is as likely to be ill-posed for this model as it is to be a hard target, and nothing measured here distinguishes the two.

Figure 12 is Figure 9 again, which is the point: the correction moved the objective and the fitted bank, and left the residual where Section 9 located it. The partial-level term reads the same fact from the other side. The model's loudest partial is its fundamental at 118 Hz, while the reference's is at 213 Hz — a partial the model does place, to within 43 cents, but sets 4.5 dB below its own loudest and

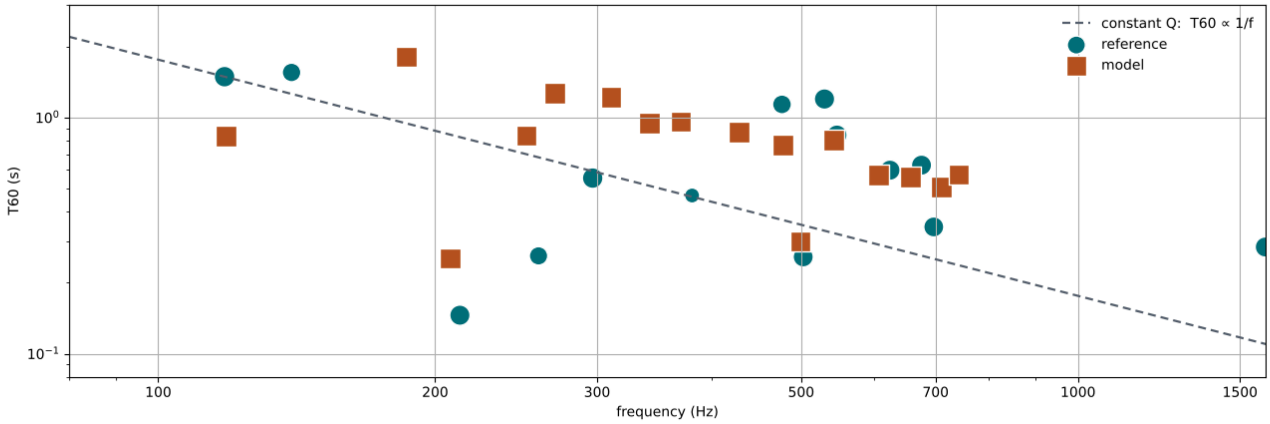

Figure 12: Ring times of the refit against the reference, drawn as Figure 9 is. The residual is the same shape and it has not softened: across 200–750 Hz the model delivers 0.5–1.3 s at all but one of its partials, where the reference scatters from 0.15 to 1.2 s, and at the fundamental it now rings 0.83 s against the reference's 1.49 — shorter than the run of Section 8 managed there. Its longest-ringing partial, 1.81 s at 186 Hz, has no counterpart in the reference at all.

rings for 0.25 s where the reference gives it 0.15. A loss law that cannot make the reference's non-monotone ring times also cannot make its partial balance, because on this recording the two are the same statement.

Coverage is where the refit has genuinely closed. Eleven of the fourteen reference partials find a counterpart within 5%, the fundamental to within 6 cents, which is what an unmatched share of 0.047 means concretely; what is missing is the 1598 Hz component and the two quiet, long-ringing partials at 140 and 530 Hz. The 0.327 spurious share is charging for five invented modes. The 476–700 Hz band — the one the reduction of Section 6 turns on — holds five fitted partials against the reference's seven, exactly what the earlier winner managed. Its density is still not reproduced.

13.4. Two notes on the seeding

The four seeded restarts converged at 26.777, 26.777, 26.777 and 26.793 cents of frequency error, which is the best seed error yet measured against a corrected target — against the 35.9–37.0 cents of Table 12, and within 7% of the 25-cent gate. The reading offered in Section 10, that no bank the product can express places its modes much better than the middle thirties on this target, is therefore **too strong as stated** and is narrowed rather than withdrawn. Whether the gate is reachable at all remains open.

The second note is a caution about the instrument that produced the first. Four seeds from four different starts agreeing to three decimal places is not what four diverse optima look like. Either the pre-solve's diversity constraint is not producing diverse banks or this is one sharp optimum every start falls into, and the report does not distinguish them — which is worth settling before `-seeded-restarts` is used to argue anything. Seeding lost again on the paired reading: best 10.970 seeded against 10.382 unseeded, mean 13.05 against 11.68, and the winning restart was unseeded for the third run running.

14. WHAT TO WATCH FOR

Eleven points generalise beyond this instrument, each learned by measurement.

An exact formula can be an unusable estimator; probe it before trusting it. Correctness and conditioning are different properties, and a derivation establishes only the first. Sweep the inputs an estimator is expected to see and look at what its correction factor does across that range: if the correction spans 100 dB, the output is a restatement of whichever input drives it, however exact the algebra. That is discoverable by measurement long before it is visible by inspection.

Loosen a guard only after the quantity it ranks on is sound. Guards and estimators interact: a tight threshold can be silently protecting a downstream computation from the region where it misbehaves, so relaxing the threshold first makes the measurement worse and blames the wrong component. Bound the estimator, then open the aperture — in that order.

Measure inter-channel lag before summing. A stereo pair of one event is one signal twice, and averaging imposes a fixed comb on every spectral quantity downstream. A low zero-lag correlation between two views of one close-miked hit is evidence that the channels are offset in time, not that there is a room on it.

Score coverage both ways, and weight the two sides equally. Either direction alone admits a degenerate optimum. Weighting the invented-partial side more heavily than the missing-partial side inverts the degeneracy rather than closing it — discarding the drum becomes the cheapest way to invent nothing, and a run under such a weighting duly converged on two partials. The pressure toward completeness and the pressure toward tidiness are one quantity seen from two sides; the asymmetry belongs in the blend, where it is already present, and not in the weights.

Read a pinned parameter as a question about the range, and answer it by measurement. A value sitting on a bound is among the most informative things an optimiser produces, so report the normalised position alongside the engineering value: the difference between "fitted to 0.276" and "fitted to the bound" should be visible without opening the bank. Then settle it by giving the search room outside the shipped box rather than by widening the box — a build-time multiplier costs nothing downstream, whereas a widened spec silently retunes every preset that stores a normalised position. Here the answer was that the bound never bound.

Distance to a mode is not distance to a heard partial. A pre-solve over analytic mode frequencies scores the distance to the nearest **mode**; the gate scores the distance to the nearest **detected partial**. A mode can sit exactly on a reference partial and never be heard — wrong radiation weight, in a pickup null, or buried under a louder neighbour — so a 1.0-cent seed does not promise a 1.0-cent frequency term, and the tens of cents the seeded fit actually reached is the measure of the gap. Mode placement is necessary for that term and nowhere near sufficient.

Let a cheap pre-solve decide whether its own output is worth using. Seeding a restart trades range for a head start, and the trade is only worth making when the head start is large. The useful part is that the same computation which produces the seed also reports how good it is, in the units the gate is written in and before any expensive evaluation — so the flag can be raised or left down on evidence rather than on hope. Here the identical mechanism gained

12% against a target it could reach to a cent and lost against one it could only reach to thirty-six, with the winning restart unseeded in both full-budget fits. Prefer a warm start whose quality you can read in advance, and treat "seeding helps" as a statement about a target rather than about a method.

A term that will not move under any intervention is a question about the term. Where the other eight terms of this distance respond to the contact model, to the damping range, to seeding, to two corrected reductions and to a converged full-budget refit, the spectral envelope has sat between 11 and 13.6 dB through all six — the largest single contributor in both fits of Table 9, and second only to partial level in Section 13. That pattern is worth reading as evidence about the measurement as much as about the model: this term is mean-removed band shape out to 12.5 kHz, and above roughly 3 kHz the model offers only its stochastic attack layer — no shell, no bearing edge, no hardware. A threshold applied over a span the model cannot structurally populate asks for something no parameter reaches, and the remedy is then to restrict the band or re-derive the threshold rather than to keep fitting against it. Whether that is the case here is not settled and is being measured; the transferable point is that immobility under independent interventions is a signal, and a per-term apparatus is what lets it be seen.

Check a cheap objective for discrimination before quoting its numbers. Reading both heads roughly doubles the mode count, and a dense enough bank is near **any** frequency by accident. Measured over 2000 random banks, the audibility-weighted frequency error has median 164.8 cents, tenth percentile 56.2 cents and best 4.1 cents, so a 1.0-cent seed is a real selection from that distribution rather than an artefact of density. An objective that cannot separate a good bank from a random one will seed noise just as confidently.

An estimator must be able to say that it did not measure. A probe placed at a fixed time on a partial that decays is a probe that eventually reads something else, and it reports that reading in the same units, on the same scale, with the same air of authority as a good one — so the failure is invisible exactly where it is worst. Two guards close it, and both are cheap: place the second probe relative to the tracked quantity's own decay rather than at a fixed offset, and return a **measured** flag beside the value so that "did not bend" and "could not be read" are different answers. Without the flag the two collapse onto zero, and a zero flows into every sum downstream as though it were evidence. Section 12 is what that cost here.

Gate per term, and confirm by ear. A distance can be monotone in quality — ranking candidates in the order a listener would — while being wrong about all of them in absolute terms. Ordering is much easier to get right than calibration, and a converged search reports the former while appearing to report the latter.

15. SCOPE AND REPRODUCIBILITY

The model, the feature extraction, the distance and the search are in `internal/physical`, `internal/physical/match` and `cmd/fit-physical` the loss multiplier of Section 9 and the pre-solve of Section 10 are flags of that command and are recorded in the report and in the checkpoint fingerprint, so neither can be mistaken for a default run. Runs are deterministic given the seed and are excluded from continuous integration, since they take minutes and require a recording the repository does not contain. That search is derivative-free, and it need not have been: this model is nearly analytic in its parameters — mode frequencies and decay rates are closed form, which is what Section 10 exploits — so gradient-based fitting with the modal parameters constrained to stay physical during training is available [17] and is not used here. The channel-alignment behaviour, the audibility weighting and the relationship between the two coverage weights are pinned by tests that need no recording.

Figure 6, Figure 8, Figure 9 and Figure 10 are drawn by `tools/paper-figures` — `just paper-figures` — from a single fit report, which carries both feature sets in full, so they are reproducible from a committed artefact rather than from the recording, which is not redistributable. That command reproduces the distance's own greedy partial matching rather than approximating it, so Figure 6 shows the pairing the score was actually computed over. Figure 7 is measured from the two channels directly and is not regenerable without them.

Figure 11 and Figure 12 are the same two renderers over the report of Section 13 — `just paper-figures-refit` — written under a distinct suffix rather than over the originals. A figure carries no record of the run that drew it, so redrawing a chapter's figures from a run that chapter does not describe is a silent way to make a paper wrong; each set is therefore named for its own run and neither recipe can overwrite the other's output.

The five figures of Section 2 come from a second committed artefact, written by `cmd/analyze-physical` — `just paper-data`, then the same `just paper-figures`. Nothing in it is rendered: the modal bank is Equation 3 and Equation 19 evaluated in closed form, and the cavity curves are the continuous-time solve of Equation 9 linearized at rest, so it depends on no audio, no recording and no seed — and, being linearized, it shows the cavity of Figure 1 and not the coupling of Section 2.5, whose effect exists only at amplitude and is reported here from time-domain measurement instead. It is regenerated deliberately, like the images, rather than diffed in continuous integration — which is the reverse of the reference fixture the model's regression tests use, and the two should not be confused. That fixture is **generated from the model**, so nothing in Section 2 is evidence of agreement with a real drum; the measured quantities it does rest on are cited individually in Table 3.

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